Signed Navier–Stokes tail budgets: forcing, low frequencies and unforced activation

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Abstract

For a smooth finite-energy forced Navier–Stokes evolution, we bound the absolute time integrals of direct force work and low-frequency feeding in the high-frequency enstrophy balance. The bounds use kinetic energy, total energy dissipation, a fixed cutoff, and derivatives of the force; they spend none of the high-frequency viscous dissipation. Consequently, for the forced singularities constructed in [1], the accumulated tail self-stretching minus even the full tail dissipation is unbounded at every fixed cutoff. The full-field obstruction is checked in Lean. The Fourier-tail extension below is an analytic proof, not a formalized projector theorem. We also bound the forcing derivative of a quartic caloric stretching functional using only energy-level velocity controls and smooth force norms. Its signed nonlinear renewal must therefore fail a uniform absorption budget on the same forced construction. A Fourier multiplier estimate separates every low-frequency parent and every low renewal output. Low stretching outputs have a separate energy-paid history bound, leaving a target with every parent and both internal frequencies high. Actual unforced whole-space solutions with unit initial energy and an initially empty tail refute an allowance uniform over the data and the prescribed cutoff. Their initial fields can be planar polarized in an arbitrarily thin Fourier belt. This does not exclude an allowance depending on the full datum or a separately chosen cutoff. A further positive estimate absorbs stretching pairs whose parents are sufficiently far above their output, uniformly over all occupied shells. The corresponding modified caloric balance retains the complete Navier–Stokes source. A second estimate absorbs sufficiently separated parents inside the nonlinear derivative slot. More generally, an isolated pair of very high inputs can be absorbed wherever it occurs among the five inputs. The remaining signed history bound is unproved; no arbitrary-data unforced regularity estimate is obtained. A separate periodic shear construction has recurrent positive caloric renewal on actual globally regular unforced flows. Its signed primitive and positive cost have quartic initial-data bounds, while its positive-part integral exceeds every fixed quartic allowance. This distinction persists after the separated-sector payments. An instantaneous stretching capacity removes these particular cycles but introduces a signed viscous defect; an exact example shows that this defect cannot simply be absorbed. For an approximately planar-polarized Fourier belt of width proportional to the square root of frequency, a further estimate pays the complete quintic renewal uniformly over all shells. It retains the axial velocity generated by pressure. Persistence of this geometry and interactions with its complement remain uncontrolled. The common-axis hypothesis is essential for that payment: two exactly planar families in separate dyadic shells, with different axes, have positive mixed renewal. A concentrated whole-space construction gives actual unforced bursts exceeding an energy-only allowance even for these initially coherent families. Their later geometry is not assumed invariant, and a full-initial-data history bound remains open.

1 An exact split and finite allowances

Let u, p, f solve

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \qquad \nabla \cdot u = 0, \qquad \nu > 0,$$

on $\mathbb{R}^3 \times [a, T)$, where $0 < L = T - a < \infty$. Assume smoothness and, on each compact slab $[a, b] \subset [a, T)$, $u \in C^1([a, b]; H^m(\mathbb{R}^3))$ and $f, \nabla p \in C([a, b]; H^m(\mathbb{R}^3))$ for every integer $m \geq 0$. These hypotheses justify the following strong energy identities and integrations by parts. Write

$$E(t) = \|u(t)\|_2^2, \quad Y(t) = \|\nabla u(t)\|_2^2 = \|\omega(t)\|_2^2, \quad \omega = \operatorname{curl} u.$$

Suppose

$$E(t) \leq M, \quad \|f(t)\|_2 \leq F_0, \quad \|\Delta f(t)\|_2 \leq F_\Delta \qquad (a \leq t < T).$$

All constants in this note are independent of the shorter endpoint t. The kinetic-energy identity gives

$$\int_a^t Y(s)\, \mathrm{d}s \leq B, \qquad B = \frac{E(a)}{2\nu} + \frac{L\sqrt{M} F_0}{\nu}. \tag{1}$$

Indeed, $E'/2 = -\nu Y + \langle u, f \rangle$ and $E(t) \geq 0$.

Fix an angular Fourier cutoff $K \geq 0$, and set

$$u_L = \mathbf{1}_{\{|\xi| < K\}}(D)u, \quad u_H = u - u_L, \quad \omega_H = \operatorname{curl} u_H, \quad S_H = \tfrac{1}{2}(\nabla u_H + \nabla u_H^{\mathsf{T}}).$$

These are orthogonal L^2 projections commuting with derivatives. Define the physical tail quantities

$$Z_H = \tfrac{1}{2}\|\omega_H\|_2^2, \qquad D_H = \|\nabla \omega_H\|_2^2 = \|\Delta u_H\|_2^2, \qquad \Pi_H = \int_{\mathbb{R}^3} \omega_H \cdot S_H \omega_H \, \mathrm{d}x.$$

In particular, Π_H uses the high field in all three factors. With $\Pi = \int \omega \cdot S\omega$, introduce

$$P_L = \langle -\Delta u_L, u \times \omega \rangle, \quad P_H = \langle -\Delta u_H, u \times \omega \rangle, \quad X_H = P_H - \Pi_H, \quad F_H = \langle -\Delta u_H, f \rangle.$$

Then $P_L + P_H = \Pi$, and the exact tail balance is

$$Z_H' = \Pi_H + X_H - \nu D_H + F_H. \tag{2}$$

The filtered velocity is not assumed to solve an autonomous Navier–Stokes equation: X_H retains every lower-frequency input.

Proposition 1. *Let $c_K = (K^3/(6\pi^2))^{1/2}$. Under the preceding assumptions,*

$$\int_a^t |F_H(s)|\, \mathrm{d}s \leq L\sqrt{M} F_\Delta, \tag{3}$$

$$\int_a^t |X_H(s)|\, \mathrm{d}s \leq 3Kc_K\sqrt{M}B + K^2 c_K M\sqrt{LB} =: C_L. \tag{4}$$

Neither bound uses a gradient supremum or tail dissipation of the velocity.

Proof. Integration by parts gives $F_H = -\langle u_H, \Delta f \rangle$, so $|F_H| \leq \sqrt{M} F_\Delta$. This proves (3).

For the unitary Fourier convention, Cauchy–Schwarz on the ball gives

$$\|\nabla u_L\|_\infty \leq Kc_K\sqrt{E}, \qquad \|\Delta u_L\|_\infty \leq K^2 c_K \sqrt{E}.$$

Here the gradient uses the Frobenius norm, and $c_K^2 = (2\pi)^{-3}|B_K|$. The vector and matrix bounds follow by Cauchy–Schwarz before taking a component sum.

For matrices define the trilinear form

$$b(A,B,C) = \sum_{i,j,k} A_{ji}B_{kj}C_{ki}, \qquad |b(A,B,C)| \le |A|_F|B|_F|C|_F.$$

The integrated stretching identity is $\Pi = -\int b(\nabla u, \nabla u, \nabla u)$, and the same identity holds for the smooth solenoidal field u_H. Put $A = \nabla u$, $H = \nabla u_H$ and $G = \nabla u_L = A - H$. The exact telescoping identity

$$b(A,A,A) - b(H,H,H) = b(G,A,A) + b(H,G,A) + b(H,H,G)$$

and $\|H\|_2 \le \|A\|_2$ imply

$$|\Pi - \Pi_H| \le 3\,\|\nabla u_L\|_\infty\, Y \le 3Kc_K\sqrt{M}\,Y. \tag{5}$$

Also

$$|P_L| \le \|\Delta u_L\|_\infty \,\|u\|_2\,\|\omega\|_2 \le K^2 c_K M\sqrt{Y}.$$

Now $X_H = \Pi - \Pi_H - P_L$. Integrate these two bounds and use (1) and $\int_a^t \sqrt{Y} \le \sqrt{LB}$. □

Corollary 2. *If a finite constant A satisfies*

$$\int_a^t (\Pi_H - \nu D_H)\,\mathrm{d}s \le A \qquad (a \le t < T), \tag{6}$$

then the complete gradient energy obeys

$$Y(t) \le 2Z_H(a) + 2A + 2C_L + 2L\sqrt{M}F_\Delta + K^2M.$$

In particular, a half-dissipation budget implies this conclusion, but the strictly smaller fraction is unnecessary for this fixed-cutoff gradient-energy implication.

Proof. Integrate (2), apply Proposition 1, and use $Y = 2Z_H + \|\nabla u_L\|_2^2 \le 2Z_H + K^2M$. □

For an unforced solution, one can take $M = E(a)$ and $B = E(a)/(2\nu)$. Thus the low-frequency allowance is controlled by the datum, viscosity, cutoff and horizon. This does not prove (6): its uniform signed high-frequency history remains the open estimate. Allowing the full dissipation simply exposes its relation to the gradient-energy continuation criterion; it is not a new coercive estimate.

2 Application to the actual forced construction

The whole-space construction in [1] has smooth compactly supported forcing, bounded kinetic energy, zero initial velocity, and unbounded speed as $t \uparrow 1$. On slabs exhausting $[1/2, 1)$, all hypotheses above hold with fixed M, F_0, F_Δ.

The module `MettapediaNS.ForcedSignedBudget` in the Lean library applies directly to the construction's `CandidateProperties`. It proves that its gradient energy cannot have a uniform bound: such a bound would control the order-two energy via the forced viscous estimate, and the H^2 embedding would bound speed. It also proves the full-field signed-production obstruction directly, with force work bounded by $6\sqrt{E_0(u)}\sqrt{E_2(f)}$ in the doubled energy convention, where $E_j(v) = \|\nabla^j v\|_2^2$ sums all ordered derivative components. The constructor theorem is imported at the pinned revision `f9e8bc5b38b6e212696e8a30e3e91517af887bbd`.

Corollary 3. *For every such candidate, every fixed finite $K \geq 0$, and every $\beta \leq 1$,*

$$\sup_{1/2 \leq t < 1} \int_{1/2}^{t} (\Pi_H(s) - \beta\nu D_H(s))\, \mathrm{d}s = +\infty.$$

Meanwhile $\int_{1/2}^{1} |X_H|$ and $\int_{1/2}^{1} |F_H|$ are finite.

Proof. For $\beta = 1$, a finite upper bound would give a uniform gradient-energy bound by Corollary 2, contradicting the preceding forced-candidate theorem. For $\beta \leq 1$, the accumulated quantity only increases because $D_H \geq 0$. The other assertions follow from Proposition 1. □

Thus the signed tail self-interaction is the term that defeats the budget; an infinite direct forcing-work allowance is not the cause. This conclusion applies to the complete constructed solution, without replacing its oscillatory corrections by a finite numerical model.

Forcing is still essential to that constructed history. With its zero datum and zero force, the kinetic-energy identity forces the smooth finite-energy solution to remain zero. In the construction the force is the smoothly extended momentum residual of the designed field (Section 10 of [1]). Removing it while keeping that field would change the equation. Restarting an unforced evolution from a later nonzero slice is a different problem, whose subsequent history is not settled by this argument. A proposed unforced budget must therefore use more than bounded forcing norms or bounded direct force work.

3 Forcing in a caloric stretching functional

The preceding obstruction can also test a functional that records stretching beyond the instantaneous enstrophy. Let

$$A = -\Delta, \qquad H_s = e^{-\nu s A}, \qquad R = \operatorname{curl} A^{-1} P, \qquad \mathcal{M}(v) = R\big((\operatorname{curl} v \cdot \nabla) v\big),$$

where P is the whole-space Leray projection. The multiplier R is used on a divergence: $\mathcal{M}(v) = R \operatorname{div}(v \otimes \operatorname{curl} v)$. Thus its low-frequency singularity causes no L^2 difficulty. Parseval's identity gives $\|R \operatorname{div} Q\|_2 \leq \|Q\|_2$, where the tensor norm is Frobenius. For solenoidal v,

$$\|\mathcal{M}(v)\|_2 \leq \|v\|_\infty \|\nabla v\|_2, \qquad \langle Av, \mathcal{M}(v) \rangle = \Pi(v). \tag{7}$$

Define the nonnegative functional

$$\mathcal{K}(u) = \frac{1}{\nu} \int_0^\infty \|\mathcal{M}(H_s u)\|_2^2 \, \mathrm{d}s. \tag{8}$$

It is finite for the smooth L^2 fields considered here: the integrand is bounded near zero and is $O(s^{-5/2})$ at infinity by heat smoothing.

The effective force in the projected velocity equation is $g = Pf$, not necessarily f itself. Suppose in addition that

$$\|g(t)\|_\infty \leq Q_0, \qquad \|\operatorname{curl} g(t)\|_\infty \leq Q_1 \qquad (a \leq t < T).$$

These constants are finite for the smooth compactly supported force in the construction: P is bounded on every H^m, Sobolev embedding controls Pf, and $\operatorname{curl} Pf = \operatorname{curl} f$. No L^∞ boundedness of P is assumed.

Proposition 4. *Put $c_h = (8\pi)^{-3/4}$. For a fixed smooth solenoidal slice u and direction g as above, with $E = \|u\|_2^2$ and $Y = \|\nabla u\|_2^2$,*

$$|D\mathcal{K}(u)[g]| \le \frac{c_h}{\nu^2}\left(\frac{32}{3}Q_0E^{3/4}Y^{3/4} + 16Q_1E^{5/4}Y^{1/4}\right). \tag{9}$$

Consequently the standing evolution satisfies, uniformly for $t < T$,

$$\int_a^t |D\mathcal{K}(u)[g]|\,\mathrm{d}s \le C_{\mathcal{K}} := \frac{c_h}{\nu^2}\left(\frac{32}{3}Q_0M^{3/4}L^{1/4}B^{3/4} + 16Q_1M^{5/4}L^{3/4}B^{1/4}\right). \tag{10}$$

This payment spends no order-two velocity dissipation.

Proof. Use the auxiliary heat parameter $r = \nu s$, and write $w_r = e^{-rA}u$, $h_r = e^{-rA}g$. Differentiating both parent slots gives

$$D\mathcal{M}(w)[h] = R\,\mathrm{div}(h \otimes \mathrm{curl}\, w + w \otimes \mathrm{curl}\, h).$$

The multiplier bound, heat contraction and (7) imply

$$|D\mathcal{K}(u)[g]| \le \frac{2}{\nu^2}\int_0^\infty \|w_r\|_\infty \|\nabla w_r\|_2 \left(Q_0\|\nabla w_r\|_2 + Q_1\sqrt{E}\right)\mathrm{d}r.$$

The Gaussian kernel has L^2 norm $c_h r^{-3/4}$, and its Fourier multiplier gives

$$\|w_r\|_\infty \le c_h r^{-3/4}\sqrt{E}, \qquad \|\nabla w_r\|_2 \le \min\{\sqrt{Y}, r^{-1/2}\sqrt{E}\}.$$

When $E, Y > 0$, split each integral at $r_* = E/Y$. Direct integration yields

$$\int_0^\infty \|w_r\|_\infty \|\nabla w_r\|_2^2\,\mathrm{d}r \le \frac{16}{3}c_hE^{3/4}Y^{3/4},$$
$$\sqrt{E}\int_0^\infty \|w_r\|_\infty \|\nabla w_r\|_2\,\mathrm{d}r \le 8c_hE^{5/4}Y^{1/4}.$$

These bounds also justify differentiation under the heat integral for the smooth slices. If E or Y vanishes, $u = 0$ as an L^2 whole-space field and the result is immediate. This proves (9). Finally apply Hölder in physical time, using $E \le M$, $\int Y \le B$, and the interval length at most L. □

Let $N(u) = -P((u\cdot\nabla)u)$ be the actual unforced velocity source, and set

$$\mathcal{G} = \frac{Y + \mathcal{K}}{2}, \qquad \mathcal{T} = D\mathcal{K}(u)[N(u)], \qquad \mathcal{W} = \nu\|Au\|_2^2 + \frac{1}{\nu}\|\mathcal{M}(u) - \nu Au\|_2^2.$$

In particular $\mathcal{G} \ge Y/2$ and $\mathcal{W} \ge 0$. Heat translation in (8) gives $D\mathcal{K}(u)[-\nu Au] = -\|\mathcal{M}(u)\|_2^2/\nu$. Combine this with the actual forced enstrophy identity and expand the square in $\mathcal{W}$ to obtain the exact balance

$$2\mathcal{G}' + \mathcal{W} = \mathcal{T} + D\mathcal{K}(u)[g] + 2\langle Au, f\rangle. \tag{11}$$

Both extra force terms have absolutely bounded physical-time integrals: their sum is bounded by $C_{\mathcal{K}} + 2L\sqrt{M}F_\Delta$.

Corollary 5. *On every forced candidate of the preceding section, for every $\beta \le 1$,*

$$\sup_{1/2\le t<1}\int_{1/2}^t (\mathcal{T} - \beta\mathcal{W})\,\mathrm{d}s = +\infty.$$

Proof. Integrating (11) expresses $\int(\mathcal{T}-\mathcal{W})$ as $2\mathcal{G}(t)-2\mathcal{G}(1/2)$ minus a uniformly bounded force integral. The candidate's gradient energy has no uniform upper bound, and $\mathcal{G}\geq Y/2$. This proves the case $\beta=1$; decreasing β only increases the integral since $\mathcal{W}\geq 0$. □

Thus the smooth force does not hide an infinite payment in the derivative of this quartic functional. Its signed nonlinear renewal is itself unbounded relative to its positive cost on the forced history. A proof of the corresponding unforced budget must exploit the zero-force evolution; estimating these explicit force terms is insufficient. The estimate does not bound $\mathcal{T}$: substituting $N(u)$ for g would require velocity-dependent supremum norms that are not controlled by the energy inequality.

4 Low parents and high-frequency caloric renewal

The caloric return admits a further frequency separation. Fix $K\geq 0$, write $v=P_{\geq K}u$, $\ell=u-v$, and retain $E=\|u\|_2^2$, $Y=\|\nabla u\|_2^2$, $D=\|\Delta u\|_2^2$. Choose a universal homogeneous Sobolev constant S such that $\|w\|_6\leq S\|\nabla w\|_2$ for real scalar, vector and matrix fields on $\mathbb{R}^3$. The constant c_K is as in Proposition 1.

Proposition 6. *If $E\leq M$, then for every $\varepsilon>0$,*

$$|\mathcal{T}(u)-\mathcal{T}(v)|\leq \varepsilon\nu D+C_2Y^2+C_{3/2}Y^{3/2}, \tag{12}$$

where all constants depend only on K,ν,M,ε and S:

$$a_K=\frac{2c_KS^3}{\nu^2},\qquad b_K=\frac{6c_KS^3}{\nu^2},\qquad C_2=a_K\sqrt{M}+\frac{3b_K^{4/3}M}{4(4\varepsilon\nu)^{1/3}},\qquad C_{3/2}=a_KKM.$$

In particular, the two last terms in (12) are at most $2(C_2Y+C_{3/2}\sqrt{Y})\mathcal{G}$. Their coefficient is integrable in physical time under the energy bounds.

Proof. We first bound the complete quartic Fourier kernel of $\mathcal{K}$. For ordered inputs a,b let $\mathcal{M}_b(a,b)=R((\operatorname{curl}a\cdot\nabla)b)$. At frequencies p,q with output $k=p+q$, its coefficient satisfies

$$|\widehat{\mathcal{M}_b}(p,q)|\leq|p|\,|\widehat{a}(p)|\,|\widehat{b}(q)|. \tag{13}$$

Indeed, $p\times\widehat{a}(p)$ is perpendicular to p, so its dot product with q equals its dot product with k. This cancels the $|k|^{-1}$ multiplier of R. The zero output is zero as well. Thus no lower bound on the output frequency is needed.

In an ordered quartic pairing of two such sources, heat integration gives the denominator $\nu^2(|p|^2+|q|^2+|r|^2+|s|^2)$. Its kernel is bounded by

$$\frac{|p|\,|r|}{\nu^2(|p|^2+|q|^2+|r|^2+|s|^2)}\leq\frac{1}{2\nu^2}. \tag{14}$$

All parent pairs and all interference between their heat rates are included.

For a real field a, let $a^{\#}$ be the real scalar field with Fourier transform $|\widehat{a}|$. It has the same L^2 norm and homogeneous Sobolev norms as a. Its physical values need not be nonnegative. The absolute ordered quartic pairing is bounded by $1/(2\nu^2)$ times the nonnegative Fourier convolution corresponding to $\int a^{\#}b^{\#}c^{\#}d^{\#}$. Applying Hölder and Sobolev after this Fourier majorization gives

$$|D\mathcal{K}(u)[h]|\leq\frac{2}{\nu^2}\left\|u^{\#}\right\|_6^3\|h\|_2\leq\frac{2S^3}{\nu^2}Y^{3/2}\|h\|_2. \tag{15}$$

There are four differentiated slots. Subtracting the derivatives at u and v telescopes each of the three other slots, giving twelve terms. Since v is a Fourier restriction of u, coefficient majorization yields

$$\begin{aligned}|(D\mathcal{K}(u) - D\mathcal{K}(v))[h]| &\le \frac{6}{\nu^2} \left\|\ell^{\#}\right\|_\infty \left\|u^{\#}\right\|_4^2 \|h\|_2 \\ &\le \frac{6c_K S^{3/2}}{\nu^2} E^{3/4} Y^{3/4} \|h\|_2 . \end{aligned} \tag{16}$$

The coefficient comparison takes place inside the Fourier convolution; it does not assert a pointwise inequality between $|v^{\#}|$ and $|u^{\#}|$. Here we used $\left\|\ell^{\#}\right\|_\infty \le c_K\sqrt{E}$ and interpolation $\left\|u^{\#}\right\|_4^2 \le S^{3/2}E^{1/4}Y^{3/4}$.

Write $\mathcal{B}(a,b) = -P((a\cdot\nabla)b)$. Then

$$N(u) - N(v) = \mathcal{B}(\ell, u) + \mathcal{B}(v, \ell), \qquad \|N(u) - N(v)\|_2 \le c_K\sqrt{EY} + Kc_K E.$$

Also Sobolev and interpolation give

$$\|N(v)\|_2 \le \|v\|_6 \|\nabla v\|_3 \le S^{3/2}Y^{3/4}D^{1/4}.$$

Only the L^2 contraction and derivative commutation of the sharp projector are used; no general L^p bound for the ball multiplier is assumed. Split the return exactly as

$$\mathcal{T}(u) - \mathcal{T}(v) = D\mathcal{K}(u)[N(u) - N(v)] + (D\mathcal{K}(u) - D\mathcal{K}(v))[N(v)].$$

Equations (15)–(16) bound these terms by

$$a_K(E^{1/2}Y^2 + KEY^{3/2}) + b_K E^{3/4}Y^{3/2}D^{1/4}.$$

Young's inequality with exponents 4 and 4/3 bounds the last term by $\varepsilon\nu D + 3b_K^{4/3}EY^2/(4(4\varepsilon\nu)^{1/3})$. Use $E \le M$ and $Y \le 2\mathcal{G}$ to finish. □

Low renewal outputs are paid separately, even when both source parents are arbitrarily high. Set

$$g_K = P_{<K}N(v), \qquad \mathcal{T}_K(u) = D\mathcal{K}(v)[P_{\ge K}N(v)].$$

The full source $N(v)$ is formed before splitting its output. Cauchy–Schwarz in the unitary Fourier convolution gives $|\widehat{N(v)}(\xi)| \le (2\pi)^{-3/2}|\xi|E$. Integrating this over the ball, with an extra $|\xi|$ for curl, proves

$$\|g_K\|_\infty \le \frac{K^4 E}{8\pi^2}, \qquad \|\operatorname{curl} g_K\|_\infty \le \frac{K^5 E}{10\pi^2}. \tag{17}$$

The derivative is the output frequency because the nonlinearity is in divergence form; this estimate assumes no bound on parent gradients. Proposition 4, applied at the slice v, therefore gives

$$\int_a^t |D\mathcal{K}(v)[g_K]|\,\mathrm{d}s \le C_{\mathrm{out}}, \tag{18}$$

where C_{out} is the right side of (10) with $Q_0 = K^4M/(8\pi^2)$ and $Q_1 = K^5M/(10\pi^2)$. Although v is not an autonomous solution, its energy and integrated gradient energy are bounded by those of u, which is all that this application uses.

Corollary 7. *Suppose $A_0 \geq 0$ and $\theta < 1$ satisfy, on every shorter prefix,*

$$\int_a^t \mathcal{T}_K(u)\,\mathrm{d}s \leq A_0 + \theta \int_a^t \mathcal{W}(u)\,\mathrm{d}s. \tag{19}$$

Choose $0 < \varepsilon < 1-\theta$, and let $C_f = C_{\mathcal{K}} + 2L\sqrt{M}F_\Delta$ for the forced equation, or $C_f = 0$ when $f = 0$. Then

$$\mathcal{G}(t) \leq \left(\mathcal{G}(a) + \frac{A_0 + C_{\mathrm{out}} + C_f}{2}\right) \exp\Big(C_2 B + C_{3/2}\sqrt{LB}\Big).$$

Proof. The exact split is $\mathcal{T}(u) = \mathcal{T}_K(u) + \mathcal{T}(u) - \mathcal{T}(v) + D\mathcal{K}(v)[g_K]$. Insert this in (11), integrate, and use Propositions 4 and 6. Since $\nu D \leq \mathcal{W}$, the remaining cost has coefficient $1 - \theta - \varepsilon > 0$ and can be dropped. Grönwall applies with coefficient $C_2 Y + C_{3/2}\sqrt{Y}$, whose integral is at most $C_2 B + C_{3/2}\sqrt{LB}$. □

For an unforced regularity argument, K and A_0 in (19) must be chosen from the full initial datum, viscosity and a prescribed finite horizon, uniformly over every shorter lifespan and prefix. That high-frequency signed-history estimate is not proved here. The result pays low parents and low outputs; it does not pay repeated high-frequency regeneration.

The same estimates also sharpen the forced test: on every actual forced candidate, for each fixed K and every $\theta < 1$,

$$\sup_{1/2 \leq t < 1} \int_{1/2}^t (\mathcal{T}_K(u) - \theta \mathcal{W}(u))\,\mathrm{d}s = +\infty.$$

A finite upper bound would give a uniform gradient-energy bound by Corollary 7, contradicting the constructed candidate. Unlike Corollary 5, this argument requires $\theta < 1$ because the low-parent estimate spends an arbitrarily small, positive part of the dissipation.

5 Low stretching outputs along the actual history

The output of $\mathcal{M}$ is distinct from the output of the nonlinear direction N. It too can be separated without restricting its parents. Fix $J \geq 0$, put $Q_- = P_{<J}$, $Q_+ = P_{\geq J}$, and define

$$\mathcal{K}_\pm(u) = \frac{1}{\nu}\int_0^\infty \|Q_\pm \mathcal{M}(H_s u)\|_2^2\,\mathrm{d}s, \qquad \mathcal{Z}_\pm(u) = \frac{1}{\nu}\|Q_\pm \mathcal{M}(u)\|_2^2, \qquad \mathcal{T}_\pm(u) = D\mathcal{K}_\pm(u)[N(u)].$$

Orthogonality gives $\mathcal{K} = \mathcal{K}_- + \mathcal{K}_+$ and $\mathcal{T} = \mathcal{T}_- + \mathcal{T}_+$. All source outputs are formed before the projection is applied.

Proposition 8. *With $c_J^2 = J^3/(6\pi^2)$,*

$$0 \leq \mathcal{K}_-(u) \leq \frac{c_J^2 E^2}{4\nu^2}, \qquad 0 \leq \mathcal{Z}_-(u) \leq \frac{c_J^2 E Y}{\nu}. \tag{20}$$

Along the actual forced evolution,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{K}_-(u) = -\mathcal{Z}_-(u) + \mathcal{T}_-(u) + D\mathcal{K}_-(u)[g]. \tag{21}$$

Consequently, uniformly for all shorter prefixes,

$$\left|\int_a^t \mathcal{T}_-(u)\,\mathrm{d}s\right| \leq \frac{c_J^2 M^2}{4\nu^2} + \frac{c_J^2 MB}{\nu} + C_{\mathcal{K}}.$$

The last term is zero for an unforced evolution. This bounds the signed primitive, not the integral of $|\mathcal{T}_-|$.

Proof. The ordered bound (13) and Cauchy–Schwarz in the full convolution give, at every nonzero output k,

$$|\widehat{\mathcal{M}(u)}(k)| \le (2\pi)^{-3/2} \int |p|\,|\widehat{u}(p)|\,|\widehat{u}(k-p)|\,\mathrm{d}p \le (2\pi)^{-3/2}\sqrt{EY}.$$

Integrating its square over $|k| < J$ proves the $\mathcal{Z}_-$ bound. No upper bound on the parent frequencies was used. If $E_s = \|H_s u\|_2^2$ and $Y_s = \|\nabla H_s u\|_2^2$, then $E_s' = -2\nu Y_s$, so

$$\mathcal{K}_-(u) \le \frac{c_J^2}{\nu}\int_0^\infty E_s Y_s\,\mathrm{d}s \le \frac{c_J^2 E^2}{4\nu^2}.$$

Heat translation gives $D\mathcal{K}_-(u)[-\nu Au] = -\mathcal{Z}_-(u)$. The actual equation therefore proves (21). Proposition 4 applies to $\mathcal{K}_-$ with unchanged constants: inserting an orthogonal output projection in both factors of its derivative can only improve the two L^2 norm bounds. Integrate the identity, use $|\mathcal{K}_-(u(t)) - \mathcal{K}_-(u(a))| \le c_J^2 M^2/(4\nu^2)$, and use $\int \mathcal{Z}_- \le c_J^2 MB/\nu$. □

For the unforced equation there is a sharper payment, independent even of the horizon. The actual kinetic-energy equality is $E' = -2\nu Y$, so (20) gives

$$\mathcal{K}_-(u(t)) + \int_a^t \mathcal{Z}_-(u)\,\mathrm{d}s \le \frac{c_J^2 E(a)^2}{4\nu^2}. \tag{22}$$

Indeed, the bound for the integral is $c_J^2(E(a)^2 - E(t)^2)/(4\nu^2)$, which complements the bound for the remaining capacity. Therefore every unforced prefix satisfies

$$-\mathcal{K}_-(u(a)) \le \int_a^t \mathcal{T}_-(u)\,\mathrm{d}s \le \frac{c_J^2 E(a)^2}{4\nu^2} - \mathcal{K}_-(u(a)).$$

This is a bound on repeated low-output renewal from the initial kinetic energy; its constant grows with the fixed output cutoff.

There is a more direct way to use this payment in the regularity target. Keep the original, full positive cost $\mathcal{W}$, and set $\mathcal{G}_+ = (Y + \mathcal{K}_+)/2 \ge Y/2$. Subtracting (21) from (11) gives the exact identity

$$2\mathcal{G}_+' + \mathcal{W} = \mathcal{T}_+ + \mathcal{Z}_- + D\mathcal{K}_+(u)[g] + 2\langle Au, f\rangle. \tag{23}$$

Thus only the spent low-output cost, bounded by $C_{\mathrm{src}} = c_J^2 MB/\nu$, is needed on the right. The endpoint $\mathcal{K}_-$ allowance is unnecessary in this identity. In the unforced case one may take $C_{\mathrm{src}} = c_J^2 E(a)^2/(4\nu^2)$ by (22).

With the input cutoff K from the preceding section, define

$$v = P_{\ge K}u, \qquad \mathcal{T}_{K,J}(u) = D\mathcal{K}_+(v)[P_{\ge K}N(v)].$$

The estimates of Proposition 6 hold with $\mathcal{K}_+$ in place of $\mathcal{K}$: its ordered quartic kernel is restricted at the common output, and has the same bound. The force estimate and the low nonlinear-output payment hold as well. Consequently, if fixed $A_0 \ge 0$ and $\theta < 1$ obey

$$\int_a^t \mathcal{T}_{K,J}(u)\,\mathrm{d}s \le A_0 + \theta\int_a^t \mathcal{W}(u)\,\mathrm{d}s \tag{24}$$

on every shorter prefix, the same Grönwall argument gives

$$\mathcal{G}_+(t) \le \left(\mathcal{G}_+(a) + \frac{A_0 + C_{\mathrm{src}} + C_{\mathrm{out}} + C_f}{2}\right)\exp\left(C_2 B + C_{3/2}\sqrt{LB}\right).$$

Here $0 < \varepsilon < 1 - \theta$ defines $C_2, C_{3/2}$ as before. The exact decomposition used in (23) is

$$\begin{aligned}\mathcal{T}_+(u) = \mathcal{T}_{K,J}(u) &+ D\mathcal{K}_+(u)[N(u) - N(v)] \\ &+ (D\mathcal{K}_+(u) - D\mathcal{K}_+(v))[N(v)] + D\mathcal{K}_+(v)[P_{<K}N(v)].\end{aligned}$$

In particular, no autonomous evolution is assigned to the filtered field v.

Taking $J = K$ leaves a quintic term whose five original velocity inputs, nonlinear-direction output, and stretching output are all above the same fixed cutoff. Its physical-time bound (24) remains open, with the same initial-data and shorter-lifespan quantifiers as (19). On the actual forced construction it fails for every fixed K, J and every $\theta < 1$: a finite upper allowance would bound $\mathcal{G}_+ \geq Y/2$, contrary to the candidate's unbounded gradient energy. This further separation pays low stretching outputs; it does not bound regeneration among the remaining high frequencies.

6 Unforced activation with unit initial energy

The allowance in (24) is permitted to depend on the full initial datum. That permission is substantive even when the initial tail is empty. In this section a subscript Q on a projector means $P_Q = \mathbf{1}_{\{|\xi| \geq Q\}}(D)$; the superscript on $\mathcal{T}^{\nu}_{Q,Q}$ or $\mathcal{W}^{\nu}$ specifies the viscosity in the functional. In particular, $\mathcal{W}^{\nu}$ is the original full-field cost, not a cost computed from $P_Q u$ alone.

Proposition 9. *For every $\nu, H, \varepsilon > 0$, $\theta \geq 0$ and finite $B \geq 0$, there are a real solenoidal Schwartz datum u_0 on $\mathbb{R}^3$, a cutoff $Q_* > 0$, and an actual smooth unforced solution from u_0 on a prefix $[0, t_*]$, $0 < t_* < H$, such that*

$$\|u_0\|_2^2 = 1, \qquad (u_0)_3 = 0, \qquad \operatorname{supp} \widehat{u}_0 \subset \{\xi : 0 < |\xi| < Q_*,\ |\xi_3| \leq \varepsilon|\xi|\},$$

but

$$\int_0^{t_*} \mathcal{T}^{\nu}_{Q_*,Q_*}(u(t))\,\mathrm{d}t > B + \theta \int_0^{t_*} \mathcal{W}^{\nu}(u(t))\,\mathrm{d}t. \tag{25}$$

Thus an energy-only allowance uniform over the datum and the prescribed cutoff is false. The cutoff varies in this construction; the proposition does not rule out a cutoff-dependent allowance, an allowance using the full datum, or the existence of a different cutoff for that datum.

Proof. We first obtain a positive first-generation coefficient on the torus, then localize it, and finally use a common local smooth time interval.

Discrete coefficient. On the 2π torus with normalized volume, write $Jk = (-k_2, k_1, 0)$. The real trigonometric field with coefficients

$$\widehat{w}(k) = i a_k Jk, \qquad (k, a_k) \in \{((3,1,1), \tfrac{1}{2}), ((1,3,1), 1), ((1,-1,3), \tfrac{1}{2})\},$$

and their reality partners has no tail above $Q^2 = 14$. For its actual first tail derivative $v_1 = P_Q N(w)$, finite Fourier convolution and integration of all parent heat rates give

$$\mathcal{T}^1_{Q,Q}(v_1) = \frac{15667851179107579904}{95414925477178125} > 0. \tag{26}$$

Every parent has squared radius 11. The internal output centers have even coordinates and hence squared radii divisible by 4; none lies on the cutoff 14. Using the cutoff 16 produces the same value.

For arbitrary thinness, use instead

$$p_m = (m, 3, 1), \quad q_m = (3, m, 1), \quad r_m = (m, 1, 3), \qquad Q_m^2 = 3m^2/2,$$

with planar coefficients ia_kJk/m, weights $(2,1,1)$ and reality partners. The parent radii are $\sqrt{m^2+10}$ and their axial ratios are at most $3/m$. Keeping all frequency labels distinct, the same finite rational calculation gives

$$\mathcal{T}^1_{Q_m,Q_m}(P_{Q_m}N(w_m)) = \frac{52224}{25m^3} - \frac{7422976}{375m^4} + O(m^{-5}). \tag{27}$$

One way to check this identity is to put $\delta = 1/m$ and represent each scaled frequency as $a+b\delta$. The scaled return has coefficients zero through order eight, followed by $52224/25$ and $-7422976/375$ at orders nine and ten. Restoring the frequencies and the first generated velocity multiplies it by m^6. Every denominator has nonzero constant term. The finitely many projector decisions are stable for small δ: their leading squared radii are integers, whereas the scaled cutoff squared is $3/2$. These are rational identities, not an assumed invariant finite-mode evolution. Fix a sufficiently large m to make the coefficient positive and the axial ratio strictly smaller than ε.

Whole-space localization. Denote the chosen periodic parent by $w_{\mathrm{per}} = J\nabla\psi$, where $J\nabla = (-\partial_2, \partial_1, 0)$, and its cutoff by Q. Choose a nonzero nonnegative Schwartz function ϕ with compact Fourier support. Such a function is the square of the real inverse Fourier transform of a nonzero real even function in C_c^∞. Set

$$w_L(x) = J\nabla(\phi(x/L)\psi(x)), \qquad v_L = P_Q N(w_L).$$

This datum is exactly solenoidal and has zero axial velocity. Its Fourier support lies in $O(L^{-1})$ neighborhoods of the parent centers; for sufficiently large L it lies strictly below Q, away from zero, and within the required Fourier belt.

The degree-ten functional $\mathcal{T}^1_{Q,Q}(v_L)$ has a finite packet expansion. Each sharp projector is evaluated near one of the finitely many internal centers, all separated from its cutoff. Each retained Leray or inverse-curl multiplier is separated from zero. Nonresonant terms vanish for large L, since the sum of their Fourier supports cannot contain zero. In a resonant term, rescale the Fourier offsets by L. The multiplier converges uniformly to its value at the centers. The parent heat frequencies are bounded away from zero, so the heat integrals are dominated by a polynomial in heat time times e^{-cs} for some $c > 0$. The compact envelope spectra then justify dominated convergence term by term. It follows that

$$\begin{aligned} L^{-3}\mathcal{T}^1_{Q,Q}(v_L) &\longrightarrow \mathcal{T}^1_{Q,Q,\mathrm{per}}(P_Q N(w_{\mathrm{per}})) \int_{\mathbb{R}^3} \phi^{10} > 0, \\ L^{-3}\,\|w_L\|_2^2 &\longrightarrow \|w_{\mathrm{per}}\|^2_{2,\mathrm{per}} \int_{\mathbb{R}^3} \phi^2 > 0. \end{aligned} \tag{28}$$

The extra velocity term from differentiating $\phi(x/L)$ is of order L^{-1} and has zero limit in these normalizations. The tenth power in (28) retains the actual quadratic generation of the child followed by the quintic renewal functional.

Fix a sufficiently large L and normalize w_L by its exact L^2 norm, obtaining a Schwartz datum w_0 with squared norm one and $P_Q w_0 = 0$. Its child $v_1 = P_Q N(w_0)$ satisfies $c = \mathcal{T}^1_{Q,Q}(v_1) > 0$; the normalization multiplies the coefficient by $\|w_L\|_2^{-10}$.

An actual common time interval. For $0 \le \delta \le 1$, let w^δ solve the unforced equation with viscosity δ and initial datum w_0. Local high-Sobolev energy estimates discard the nonpositive viscous term, giving a common smooth interval and uniform high-Sobolev bounds. Applying the equation twice also bounds two time derivatives in H^2, uniformly in δ. Since the initial tail vanishes,

$$P_Q w^\delta(\tau) = \tau v_1 + O(\tau^2) \quad \text{in } H^2,$$

uniformly for $\delta \in [0,1]$. The quartic kernel bound above gives $|D\mathcal{K}_+(v)[h]| \le C\,\|\nabla v\|_2^3\,\|h\|_2$ at viscosity one, and $N : H^2 \to L^2$ is continuous. Hence the quintic $\mathcal{T}^1_{Q,Q}$ is continuous on H^2, with bounded multilinear coefficients there, and

$$\mathcal{T}^1_{Q,Q}(w^\delta(\tau)) = c\tau^5 + O(\tau^6)$$

uniformly in δ. Choose a sufficiently small fixed $\tau_* > 0$. There are $\kappa, C_0 > 0$, independent of δ, such that

$$\int_0^{\tau_*} \mathcal{T}^1_{Q,Q}(w^\delta)\,\mathrm{d}\tau \ge \kappa, \tag{29}$$

$$\int_0^{\tau_*} \left[\delta^2 \left\|Aw^\delta\right\|_2^2 + \left\|M(w^\delta) - \delta A w^\delta\right\|_2^2\right] \mathrm{d}\tau \le C_0. \tag{30}$$

Here $A = -\Delta$. The second assertion follows directly from the common H^2 bound and $\|M(w)\|_2 \le \|w\|_\infty \|\operatorname{curl} w\|_2$. The time τ_* is fixed only after m, L, w_0 have been fixed; no numerical value or uniformity in the thinness parameter is asserted.

Energy-preserving concentration. For sufficiently large λ, set

$$\delta = \nu/\sqrt{\lambda}, \quad u_\lambda(t,x) = \lambda^{3/2} w^\delta(\lambda^{5/2}t, \lambda x), \quad Q_\lambda = \lambda Q, \quad t_\lambda = \tau_*/\lambda^{5/2}.$$

This is the actual unforced equation with physical viscosity ν; its initial squared L^2 norm is one. Its initial tail is empty above Q_λ, and the initial polarization and belt condition persist under this spatial rescaling. Keeping the spatial Jacobian and the physical-time factor gives the exact identities

$$\int_0^{t_\lambda} \mathcal{T}^\nu_{Q_\lambda,Q_\lambda}(u_\lambda)\,\mathrm{d}t = \frac{\lambda^3}{\nu^2}\int_0^{\tau_*} \mathcal{T}^1_{Q,Q}(w^\delta)\,\mathrm{d}\tau, \tag{31}$$

$$\int_0^{t_\lambda} \mathcal{W}^\nu(u_\lambda)\,\mathrm{d}t = \frac{\lambda^{5/2}}{\nu}\int_0^{\tau_*} \left[\delta^2 \left\|Aw^\delta\right\|_2^2 + \left\|M(w^\delta) - \delta A w^\delta\right\|_2^2\right] \mathrm{d}\tau. \tag{32}$$

In particular, the square in the full cost is retained. By (29)–(30), the excess in (25) is at least

$$\frac{\kappa\lambda^3}{\nu^2} - \frac{\theta C_0 \lambda^{5/2}}{\nu} \longrightarrow +\infty.$$

Also $t_\lambda \to 0$, so the prefix lies within the prescribed horizon. This proves the proposition. □

This obstruction concerns a family of different data and different prescribed cutoffs. It neither produces a singularity for one datum nor contradicts (24) with its full-data allowance and existential cutoff. In this family the full initial caloric resource already scales as λ^3/ν^2; retaining it can pay the finite activation that an energy-only or empty-tail-only allowance misses.

7 Absorbing separated parents across all output scales

There is a sector of the stretching source that can be paid without an angular or sign assumption. Fix integers $J \ge 0$, $r \ge 3$, and put $K = 2^J$. For an output $k \in \Omega_j := \{2^j \le |k| < 2^{j+1}\}$, $j \ge J$, write $c_j = \lceil 3j/2 \rceil$. An ordered pair $p + q = k$ belongs to the separated sector if

$$n := \lfloor \log_2 \max\{|p|, |q|\} \rfloor \ge c_j + r.$$

Let $\mathcal{M}_{\mathrm{far}}$ retain these contributions in the ordered Fourier formula for $\mathcal{M}$, before summing the pairs, and define

$$\mathcal{M}_{\mathrm{lo}} = P_{<K}\mathcal{M}, \qquad \mathcal{M}_{\mathrm{keep}} = \mathcal{M} - \mathcal{M}_{\mathrm{lo}} - \mathcal{M}_{\mathrm{far}}.$$

These are fixed bilinear multipliers, with a mask symmetric in the two parent frequencies. Neither the velocity nor its evolution is truncated. In particular, $\mathcal{M}_{\mathrm{keep}}$ is not an orthogonal projection of the already summed field $\mathcal{M}$.

Proposition 10. *For every smooth solenoidal whole-space field, with the unitary Fourier convention and $E = \|u\|_2^2$, $Y = \|\nabla u\|_2^2$, $D = \|\Delta u\|_2^2$,*

$$\|\mathcal{M}_{\mathrm{far}}(u)\|_2 \le 2\sqrt{17}\,2^{-r}\sqrt{ED} \le 10\,2^{-r}\sqrt{ED}. \tag{33}$$

The bound is independent of J and the number of occupied shells. Its auxiliary heat capacity satisfies

$$\mathcal{K}_{\mathrm{far}}(u) := \frac{1}{\nu}\int_0^\infty \|\mathcal{M}_{\mathrm{far}}(H_s u)\|_2^2 \,\mathrm{d}s \le \frac{50\,2^{-2r}}{\nu^2} EY. \tag{34}$$

Proof. Since $n \ge j+3$, both parents belong to $B_n = \{2^{n-1} \le |\xi| < 2^{n+1}\}$. Write $E_n = \int_{B_n} |\widehat{u}|^2$ and $d_n = 2^{2n}E_n$. The ordered bound (13), followed by Cauchy–Schwarz in the parent frequency, gives

$$|\widehat{\mathcal{M}_{j,n}}(k)| \le (2\pi)^{-3/2} 2^{n+1} E_n \qquad (k \in \Omega_j).$$

As $|\Omega_j| = (28\pi/3)2^{3j}$,

$$\left\| P_{\Omega_j}\mathcal{M}_{\mathrm{far}} \right\|_2 \le 2\sqrt{\frac{7}{6\pi^2}} \sum_{n \ge c_j + r} 2^{3j/2-n} d_n \le \sum_n A_{jn} d_n, \qquad A_{jn} = 2^{c_j - n}\mathbf{1}_{\{n \ge c_j + r\}}.$$

Every row sum of A is 2^{1-r}. Every column sum is at most 2^{1-r}, since the integers c_j strictly increase. Applying weighted Cauchy–Schwarz to each row and then the column bound proves

$$\sum_j \left(\sum_n A_{jn} d_n \right)^2 \le 2^{2-2r} \sum_n d_n^2.$$

The argument first applies to finite submatrices, and then passes to the increasing limit of their nonnegative sums. This summation avoids paying the same high-frequency tail separately at every output shell.

Each frequency of dyadic index ℓ belongs to just B_ℓ and $B_{\ell+1}$. Their fourth-order weights sum to $2^{4\ell} + 2^{4\ell+4} \le 17|\xi|^4$. Consequently

$$\sum_n d_n^2 = \sum_n 2^{4n} E_n^2 \le E \sum_n 2^{4n} E_n \le 17 ED.$$

The output shells are orthogonal, proving (33). For (34), apply that bound at $H_s u$ and use $E(H_s u) \le E$ and $\int_0^\infty D(H_s u)\,\mathrm{d}s = Y/(2\nu)$. □

Fix an energy bound $E \le M$, choose $0 < \eta < 1/2$, and choose r so that $10\,2^{-r}\sqrt{M} \le \eta\nu$. Then

$$2|\langle Au, \mathcal{M}_{\mathrm{far}}\rangle| \le 2\eta\nu D, \qquad \mathcal{K}_{\mathrm{far}} \le \eta^2 Y/2. \tag{35}$$

The separation is fixed from M, ν, η; it is not adjusted from a future solution norm.

To use this source payment, change the auxiliary capacity and its positive cost explicitly:

$$\mathcal{K}_{\mathrm{keep}}(u)=\frac{1}{\nu}\int_0^\infty \|\mathcal{M}_{\mathrm{keep}}(H_s u)\|_2^2\,\mathrm{d}s, \qquad \mathcal{G}_{\mathrm{keep}}=(Y+\mathcal{K}_{\mathrm{keep}})/2,$$
$$\mathcal{W}_{\mathrm{keep}}=\nu D+\frac{1}{\nu}\|\mathcal{M}_{\mathrm{keep}}-\nu Au\|_2^2, \qquad \mathcal{T}_{\mathrm{keep}}=D\mathcal{K}_{\mathrm{keep}}(u)[N(u)].$$

In particular $\mathcal{G}_{\mathrm{keep}}\ge Y/2$ and $\mathcal{W}_{\mathrm{keep}}\ge \nu D$. This is a modified cost, not an identification with the original $\mathcal{W}$. Heat translation for the fixed mask, the full enstrophy identity, and expansion of the square give, along the actual equation $u_t=N(u)-\nu Au+g$,

$$\begin{aligned}2\mathcal{G}'_{\mathrm{keep}}+\mathcal{W}_{\mathrm{keep}}=\mathcal{T}_{\mathrm{keep}}+D\mathcal{K}_{\mathrm{keep}}(u)[g]+2\langle Au,g\rangle\\ +2\langle Au,\mathcal{M}_{\mathrm{lo}}\rangle+2\langle Au,\mathcal{M}_{\mathrm{far}}\rangle.\end{aligned}\tag{36}$$

There is no claim that $\mathcal{K}$ is the sum of the three capacities: the retained and separated pairs can interfere at the same output. The identity uses the complete $N(u)$ in the derivative direction.

The low-output estimate $\|\mathcal{M}_{\mathrm{lo}}\|_2\le c_K\sqrt{EY}$ implies

$$|\langle Au,\mathcal{M}_{\mathrm{lo}}\rangle|\le K^2c_KE\sqrt{Y}, \qquad 2\int_a^t|\langle Au,\mathcal{M}_{\mathrm{lo}}\rangle|\,\mathrm{d}s\le 2K^2c_KM\sqrt{LB}=:C_{\mathrm{lo}}.$$

Thus the two last terms of (36) cost a finite allowance and a fraction 2η of $\mathcal{W}_{\mathrm{keep}}$.

For completeness, the other payments must respect the parent mask. Every retained ordered symbol still obeys (13); the two masks in the quartic pairing are bounded by one. Hence Proposition 6 applies to $\mathcal{K}_{\mathrm{keep}}$ with exactly the same constants. For the forcing estimate use the Fourier moments

$$Q_\ell^{\#}=(2\pi)^{-3/2}\int_{\mathbb{R}^3}|\xi|^\ell|\widehat{g}(\xi)|\,\mathrm{d}\xi, \qquad \ell=0,1,$$

rather than asserting an L^∞ bound for the masked bilinear operator. Fourier convolution and Young's inequality give, with $w=H_su$,

$$\|\mathcal{M}_{\mathrm{keep}}(w)\|_2\le\left\|w^{\#}\right\|_\infty\|\nabla w\|_2, \qquad \|D\mathcal{M}_{\mathrm{keep}}(w)[H_sg]\|_2\le Q_0^{\#}\|\nabla w\|_2+Q_1^{\#}\sqrt{E}.$$

Here $\left\|w^{\#}\right\|_\infty=(2\pi)^{-3/2}\int|\widehat{w}|$; the physical values of $w^{\#}$ need not be nonnegative. The heat-splitting proof of Proposition 4 now applies with $Q_0^{\#},Q_1^{\#}$ in place of Q_0,Q_1. These moments are bounded by the H^2 and H^3 norms of g, respectively. They are finite uniformly in time for the force in the construction.

For $v=P_{\ge K}u$ and $g_K=P_{<K}N(v)$, the Fourier proof of (17) bounds these same moments:

$$Q_0^{\#}(g_K)\le K^4E/(8\pi^2), \qquad Q_1^{\#}(g_K)\le K^5E/(10\pi^2).$$

It therefore also pays the low nonlinear output of the retained capacity. Let C_{out} be the resulting time-integral allowance, and let C_f include the two explicit force terms of (36). They are finite under the preceding energy and force assumptions.

The remaining signed term is exactly

$$\mathcal{T}_{\mathrm{core}}(u)=D\mathcal{K}_{\mathrm{keep}}(v)[P_{\ge K}N(v)], \qquad v=P_{\ge K}u.$$

Its five original velocity inputs, nonlinear-direction output and stretching output are all above K. The outer stretching pairs additionally satisfy

$$\max\{|p|,|q|\}<2^{c_j+r}\le\sqrt{2}\,2^r|p+q|^{3/2} \qquad (p+q\in\Omega_j).$$

This restriction is imposed on the two slots of each stretching source. In a differentiated slot, $N(v)$ has two further parents; the displayed restriction imposes no corresponding comparability on those two parents. Their transfer and the signed time history remain to be controlled.

If constants $A_0 \geq 0$, $\theta < 1 - 2\eta$ obey

$$\int_a^t \mathcal{T}_{\mathrm{core}}(u)\,\mathrm{d}s \leq A_0 + \theta \int_a^t \mathcal{W}_{\mathrm{keep}}(u)\,\mathrm{d}s \tag{37}$$

on every shorter prefix, choose $0 < \varepsilon < 1 - 2\eta - \theta$ in Proposition 6. The same exact low-parent decomposition and integral Grönwall argument then yield

$$\mathcal{G}_{\mathrm{keep}}(t) \leq \left(\mathcal{G}_{\mathrm{keep}}(a) + \frac{A_0 + C_{\mathrm{lo}} + C_{\mathrm{out}} + C_f}{2}\right) \exp\Bigl(C_2 B + C_{3/2}\sqrt{LB}\Bigr)\,.$$

The target (37) remains open. For regularity its cutoffs, separation and allowances must be fixed from the full initial datum, viscosity and horizon, uniformly over shorter solution intervals. The actual forced singularities violate this target: otherwise the displayed bound would control their unbounded gradient energy.

The positive generated-child example (26) is retained at $J = 2$, $r = 3$ with exactly the same coefficient. Thus the absorption does not remove all positive renewal or make a sign claim about the remaining term. It pays one frequency sector uniformly, leaving a smaller physical-time problem.

The change of capacity does not impose a stronger boundedness requirement on the solution. Indeed, the triangle inequality in the auxiliary $L^2(\,\mathrm{d}s/\nu; L^2_x)$ space, the low-output capacity bound, and (35) give, when $E \leq M$ and $\eta < 1/2$,

$$\mathcal{G}_{\mathrm{keep}} \leq 3\mathcal{G} + \frac{3c_K^2 M^2}{8\nu^2}, \qquad \mathcal{G} \leq 3\mathcal{G}_{\mathrm{keep}} + \frac{3c_K^2 M^2}{8\nu^2}.$$

For example, $\mathcal{K}_{\mathrm{keep}} \leq 3\mathcal{K} + 3\mathcal{K}_{\mathrm{lo}} + 3\mathcal{K}_{\mathrm{far}}$, with $\mathcal{K}_{\mathrm{lo}} \leq c_K^2 M^2/(4\nu^2)$ and $\mathcal{K}_{\mathrm{far}} \leq \eta^2 Y/2$, proves the first inequality; the reversed source decomposition proves the second. These compare the gradient-controlling functionals, not their different positive costs.

7.1 Separated parents inside the nonlinear direction

The two parents inside the $N(v)$ slot admit another uniform payment. Expand $\mathcal{T}_{\mathrm{core}}$ in its ordered quartic derivative slots and then expand the actual quadratic source in the differentiated slot. There are five original velocity inputs. For each such term let $\zeta_1, \zeta_2, \zeta_3$ be the three undifferentiated input frequencies, and let p, q be the two parents inside $N(v)$, with output $\xi = p + q$. Frequency conservation expresses ξ as a signed sum of the three ζ_i, so $|\xi| \leq \sum_i |\zeta_i|$. Write

$$m = \left\lfloor \log_2 \max_i |\zeta_i| \right\rfloor, \qquad n = \lfloor \log_2 \max\{|p|, |q|\} \rfloor\,, \qquad b_m = \lceil 11m/8 \rceil.$$

Here $m \geq J \geq 0$, since all original inputs are above K. Fix an integer $\rho \geq 4$ and let $\mathcal{T}_{\mathrm{inner}}$ be the part of this expansion with $n \geq b_m + \rho$, retaining all the previous masks. This is a partition of the quintic return, not a replacement for the equation or the differentiated capacity.

Proposition 11. *Uniformly in the fixed cutoffs and outer separation,*

$$|\mathcal{T}_{\mathrm{inner}}(u)| \leq \frac{20\,2^{-4\rho}}{\nu^2} E^{3/2} D. \tag{38}$$

Thus, if $E \leq M$, for any $\zeta > 0$ choosing ρ from $20\,2^{-4\rho} M^{3/2} \leq \zeta\nu^3$ pays this entire term by $\zeta\nu D$.

Proof. Put $R_m = 2^{m+1}$. The three undifferentiated frequencies lie in the ball of radius R_m, and $|\xi| < 3R_m$. Since $n \geq m+4$, both inner parents lie in $B_n = \{2^{n-1} \leq |\xi'| < 2^{n+1}\}$: their difference in magnitude is at most $|\xi| < 6\,2^m \leq (3/8)2^n$. Define $E_n = \int_{B_n} |\widehat{v}|^2$. The divergence form of N bounds its part with maximal parent shell n by

$$|\widehat{N_n(v)}(\xi)| \leq (2\pi)^{-3/2}|\xi|E_n.$$

Each ordered quartic heat kernel is bounded by $1/(2\nu^2)$ as in (14); the previous masks only decrease its absolute majorant. Summing the four derivative slots, then integrating the three undifferentiated Fourier magnitudes, bounds the contribution at (m,n) by

$$\begin{aligned}\frac{2}{\nu^2}(3R_m)E_n\left((2\pi)^{-3/2}\int_{|\xi'|<R_m}|\widehat{v}(\xi')|\,\mathrm{d}\xi'\right)^3 &\leq \frac{6c_1^3}{\nu^2}R_m^{11/2}E^{3/2}E_n \\ &\leq \frac{1}{\nu^2}2^{11m/2}E^{3/2}E_n,\end{aligned}$$

where $c_1 = (6\pi^2)^{-1/2}$ and $6c_1^3\,2^{11/2} < 1$. The three convolution normalization factors are included. The N slot is heat-evolved at frequency ξ: one must not replace its heat rate $|\xi|^2$ by $|p|^2 + |q|^2$.

Set $e_n = 2^{4n}E_n$. Since b_m strictly increases,

$$\sum_{\{m:\ b_m+\rho\leq n\}} 2^{11m/2-4n} \leq \sum_{\{m:\ b_m+\rho\leq n\}} 2^{4b_m-4n} \leq \frac{16}{15}\,2^{-4\rho}.$$

Finally $\sum_n e_n \leq 17D$, by the same enlarged-band overlap calculation as before. Summing the nonnegative majorants first over m and then n gives constant $272/15 < 20$ in (38). These estimates also justify the regrouping of the absolutely convergent separated sector. □

Define the remaining return by $\mathcal{T}_{\mathrm{rem}} = \mathcal{T}_{\mathrm{core}} - \mathcal{T}_{\mathrm{inner}}$. It suffices in (37) to bound $\mathcal{T}_{\mathrm{rem}}$ instead, with $\theta < 1 - 2\eta - \zeta$ and $0 < \varepsilon < 1 - 2\eta - \zeta - \theta$. The same bound for $\mathcal{G}_{\mathrm{keep}}$ then follows. The new parameter ρ is fixed from the energy bound, viscosity and desired absorbed fraction, uniformly over the physical history. No sign assumption is used in either separated sector. This still leaves nontrivial signed renewal: the positive coefficient (26) is unchanged after this second removal at $J = 2$, $r = 3$, $\rho = 4$.

This refinement constrains very unequal parent frequencies. It does not establish that all remaining frequencies are comparable, that a pancake geometry persists, or that positive renewal cannot repeat. The all-prefix bound for $\mathcal{T}_{\mathrm{rem}}$ is still an open estimate with full-initial-data quantifiers. It also fails on the forced singularity construction, for the same continuation reason.

7.2 An isolated pair among all five inputs

The preceding inner-parent estimate distinguishes two particular slots. A further payment applies regardless of the position of the two highest frequencies. In each ordered term of $\mathcal{T}_{\mathrm{core}}$, list the dyadic indices of its five original velocity inputs in decreasing order. Let n be the largest and m the third largest. Fix an integer $\sigma \geq 4$. Among the terms not already assigned to $\mathcal{T}_{\mathrm{inner}}$, let $\mathcal{T}_{\mathrm{pair}}$ retain those with

$$n \geq \lceil 3m/2 \rceil + \sigma.$$

This order of selection makes the sectors disjoint, including when the two highest frequencies both occur inside N.

Proposition 12. *With the same whole-space conventions,*

$$|\mathcal{T}_{\mathrm{pair}}(u)| \le \frac{100\,2^{-3\sigma}}{\nu^2} E^{3/2} D. \tag{39}$$

The constant is independent of the number of occupied shells, all the previous cutoffs, and the positions of the two highest inputs.

Proof. Put $R_m = 2^{m+1}$. The three smaller frequencies lie in the ball of radius R_m. Frequency conservation bounds the difference in the magnitudes of the two larger frequencies by $3R_m$. Since $n \ge m+4$, both of the latter frequencies belong to $B_n = \{2^{n-1} \le |\xi| < 2^{n+1}\}$. As before put $E_n = \int_{B_n} |\widehat{v}|^2$.

There are four differentiated quartic slots and at most $\binom{5}{2} = 10$ choices of the two highest positions. Each quartic kernel has absolute bound $1/(2\nu^2)$. The divergence-form symbol of the inserted N is bounded by its output frequency, hence by 2^{n+2}. For each choice of positions, integrate one high frequency using Cauchy–Schwarz against the other, whose value is fixed by the three smaller frequencies. This costs at most E_n. The three other integrals are bounded by their Fourier L^1 norms on the ball. All three convolution normalizations are included, so the contribution at (m,n) is at most

$$\begin{aligned}\frac{20}{\nu^2} 2^{n+2} E_n \left((2\pi)^{-3/2} \int_{|\xi|<R_m} |\widehat{v}(\xi)|\,\mathrm{d}\xi \right)^3 &\le \frac{80 c_1^3\, 2^{9/2}}{\nu^2} 2^{9m/2+n} E^{3/2} E_n \\ &\le \frac{5}{\nu^2} 2^{9m/2+n} E^{3/2} E_n.\end{aligned}$$

Here $c_1 = (6\pi^2)^{-1/2}$ as before, and $80 c_1^3\, 2^{9/2} < 5$. This is an absolute majorant; excluding $\mathcal{T}_{\mathrm{inner}}$ or retaining any previous mask cannot increase it.

Write $d_m = \lceil 3m/2 \rceil$ and $e_n = 2^{4n} E_n$. The increasing integer sequence d_m gives the column bound

$$\sum_{\{m:\ d_m+\sigma \le n\}} 2^{9m/2-3n} \le \sum_{\{m:\ d_m+\sigma \le n\}} 2^{3d_m-3n} \le \frac{8}{7}\, 2^{-3\sigma}.$$

Summing these nonnegative bounds, then using $\sum_n e_n \le 17D$, gives constant $680/7 < 100$ in (39). Finite frequency restrictions followed by monotone convergence justify the summation. □

For $E \le M$ choose σ from $100\,2^{-3\sigma} M^{3/2} \le \chi\nu^3$, where $\chi > 0$. Then $|\mathcal{T}_{\mathrm{pair}}| \le \chi\nu D$. Define the remaining term

$$\mathcal{T}_{\mathrm{coupled}} = \mathcal{T}_{\mathrm{core}} - \mathcal{T}_{\mathrm{inner}} - \mathcal{T}_{\mathrm{pair}}.$$

An all-prefix bound for its signed integral by $A_0 + \theta \int \mathcal{W}_{\mathrm{keep}}$, with $\theta < 1 - 2\eta - \zeta - \chi$, suffices for the same Grönwall conclusion; take $0 < \varepsilon < 1 - 2\eta - \zeta - \chi - \theta$. Every separation and allowance must still be fixed from the initial datum, viscosity and horizon before the shorter endpoints.

Every retained ordered term has $n < \lceil 3m/2 \rceil + \sigma$. If $\lambda_{\max}$ and $\lambda_{(3)}$ are the largest and third-largest original input magnitudes, then

$$\lambda_{\max} < 2^{\sigma+1/2} \lambda_{(3)}^{3/2}.$$

Thus at least three inputs must exceed a fixed multiple of $\lambda_{\max}^{2/3}$. This is a restriction on each remaining five-input interaction, not comparability of all its scales or a persistence theorem. The positive generated-child coefficient (26) lies wholly in this remaining sector at $J = 2$, $r = 3$, $\rho = \sigma = 4$. Its signed physical-time budget remains unproved.

A separate pressure-free calibration clarifies what this return measures. On the normalized 2π torus take

$$u = (\sin y,\, 0,\, \sin x + \sin(x+y)).$$

Its actual Euler direction is $N(u) = (0, 0, -\sin y\,(\cos x + \cos(x+y)))$, with zero pressure. At $J = 0$, $r = 3$, $\rho = \sigma = 4$, exact Fourier calculation gives

$$\mathcal{T}_{\text{inner}} = \mathcal{T}_{\text{pair}} = 0, \qquad \mathcal{T}_{\text{coupled}} = \frac{161}{19200\nu^2} > 0, \qquad \Pi(u) = -\frac{1}{4}, \qquad D\Big(\int |u|^4\Big)\,[N(u)] = 0.$$

Replacing $\sin(x+y)$ by its negative reverses the two nonzero signs. The actual unforced viscous evolution is a heat-evolved shear and a passive scalar: $U_t = \nu U_{yy}$ and $\Theta_t + U\Theta_x = \nu\Delta\Theta$. It remains smooth for all time. Thus positive coupled renewal need not mean positive physical enstrophy production or positive pressure work. This calibration does not refute a full-datum history allowance.

8 Recurrent caloric renewal in a regular shear

In this section only, space is the normalized 2π torus. The symbols $\mathcal{K}^\nu, \mathcal{T}^\nu, \mathcal{W}^\nu$ denote the periodic counterparts of the full, unmasked functionals. The following example concerns repeated renewal, rather than an initial derivative or one burst.

Proposition 13. *There is a fixed smooth, real, mean-zero solenoidal periodic field v_0 such that the globally smooth unforced solutions u^α with initial data αv_0 have the following property. For every fixed $\nu, H > 0$, there are $c_\nu, C_\nu > 0$ and times $t_\alpha = \gamma \log\alpha/\alpha \le H$ for all sufficiently large α, with*

$$\int_0^{t_\alpha} (\mathcal{T}^\nu(u^\alpha))_+ \,\mathrm{d}t \ge c_\nu \alpha^4 \log\alpha, \tag{40}$$

$$\left|\int_0^t \mathcal{T}^\nu(u^\alpha)\,\mathrm{d}s\right| \le C_\nu\alpha^4, \qquad \int_0^t \mathcal{W}^\nu(u^\alpha)\,\mathrm{d}s \le C_\nu(\alpha^2+\alpha^4) \quad (t \ge 0). \tag{41}$$

In particular,

$$\frac{\displaystyle\int_0^{t_\alpha} (\mathcal{T}^\nu(u^\alpha))_+ \,\mathrm{d}t}{\displaystyle Y(u_0^\alpha) + \nu^{-2}\,\|u_0^\alpha\|_4^4 + \mathcal{K}^\nu(u_0^\alpha) + \int_0^{t_\alpha} \mathcal{W}^\nu(u^\alpha)\,\mathrm{d}t} \longrightarrow +\infty.$$

The same exclusion holds after adding any fixed nonnegative initial functional homogeneous of degree at most four and finite on v_0.

Proof. We first construct a smooth inviscid periodic reference with nonconstant caloric capacity. Let $U_*(y) = \operatorname{sgn}(\cos y)$ and $\phi_{\pm,*} = (1 \pm U_*)/2$. These nonsmooth functions are used only to estimate a heat pairing; they will not be used as Euler or Navier–Stokes initial data.

For a shear U and a scalar profile ϕ define its unit-viscosity caloric image $\Phi(U,\phi)$ in $L^2(\,\mathrm{d}r; L^2_{x,y})$ by

$$\Phi(U,\phi)(r,x,y) = e^{-r}\cos x\,(1-\partial_y^2)^{-1}\partial_y\big((e^{r\partial_y^2}U)' e^{r\partial_y^2}\phi\big).$$

This is the third component of $\mathcal{M}$ for the heat-smoothed velocity $(U(y), 0, \phi(y)\sin x)$; the other components vanish. The elementary multiplier bounds are

$$\left\|\partial_y(1-\partial_y^2)^{-1}\right\|_{2\to2} \le \frac{1}{2}, \qquad \left\|\partial_y^2(1-\partial_y^2)^{-1}\right\|_{2\to2} \le 1.$$

Heat contraction and its energy identity give

$$\|\Phi(U,\phi)\| \le \frac{1}{4}\,\|U\|_2\,\|\phi\|_\infty\,, \qquad \|\Phi(U,\phi)\| \le \frac{3}{4}\,\|U\|_\infty\,\|\phi\|_2\,. \tag{42}$$

For the second bound write $U_r'\phi_r = (U_r\phi_r)' - U_r\phi_r'$. The two contributions to the image norm, including the cosine factor, are at most $\|U\|_\infty\,\|\phi\|_2/2$ and $\|U\|_\infty\,\|\phi\|_2/4$. Here one uses

$$\int_0^\infty e^{-2r}\,\mathrm{d}r = \frac{1}{2}, \qquad \int_0^\infty \|\phi_r'\|_2^2\,\mathrm{d}r \le \frac{1}{2}\,\|\phi\|_2^2\,.$$

The first bound in (42) uses the same heat energy inequality for U_r'. These estimates also define the images for the step profiles by L^2 approximation.

Put $K_0 = \|\Phi(U_*,1)\|^2$ and $K_1 = \|\Phi(U_*,U_*)\|^2$. The first Fourier coefficients of U_*, at $n=\pm1$, are $2/\pi$. Their contribution alone gives

$$K_0 \ge \frac{1}{4\pi^2} > \frac{1}{40}.$$

For $U_r = e^{r\partial_y^2}U_*$ put $a(r) = \|U_r\|_2^2$. The bounds $|U_r| \le 1$ and $a(r) \le e^{-2r}$ imply

$$\left\|U_r^2 - \textstyle\int U_r^2\right\|_2^2 \le a(r)(1-a(r)) \le \min\{1/4, e^{-2r}\}.$$

Since $U_r'U_r = (U_r^2)'/2$, the second multiplier bound yields

$$K_1 \le \frac{1}{8}\int_0^\infty e^{-2r}\min\{1/4, e^{-2r}\}\,\mathrm{d}r = \frac{7}{512}.$$

The integral is split at $r = \log 2$. The two images $\Phi(U_*,1)$ and $\Phi(U_*,U_*)$ have disjoint odd and even y-frequencies. Consequently their packet cross pairing satisfies

$$C_* := \langle \Phi(U_*,\phi_{+,*}), \Phi(U_*,\phi_{-,*})\rangle = \frac{K_0-K_1}{4} > \frac{29}{10240}.$$

Choose smooth real U,ϕ_+,ϕ_-, agreeing with the step profiles outside a set of normalized y-measure $d \le 2^{-20}$, such that

$$|U| \le 1, \quad 0 \le \phi_\pm \le 1, \quad U\phi_\pm = \pm\phi_\pm, \quad U'\phi_\pm = 0.$$

They can be obtained by symmetric smooth transitions near the two jump points, with the packet cutoffs supported strictly inside the flat regions. Choose $U(y+\pi) = -U(y)$ and $\phi_-(y) = \phi_+(y+\pi)$, so U has mean zero and the packet supports are disjoint. Their L^2 errors are at most $2\sqrt{d}, \sqrt{d}, \sqrt{d}$. By bilinearity and (42), each packet image changes by at most $5\sqrt{d}/4$, and every image has norm at most $1/4$. Thus the smooth cross pairing C satisfies

$$|C - C_*| \le \frac{5}{8}\sqrt{d} \le \frac{5}{8192}, \qquad C > \frac{91}{40960} > \frac{1}{512}.$$

The actual Euler reference is

$$v^0(\tau) = (U(y), 0, \phi_+(y)\sin(x-\tau) + \phi_-(y)\sin(x+\tau)).$$

It is smooth, 2π-periodic in τ, solves Euler with zero pressure, and has $(\operatorname{curl} v^0\cdot\nabla)v^0 = 0$. Its Sobolev norms are uniformly bounded. Nevertheless, if $K_\pm = \|\Phi(U,\phi_\pm)\|^2$,

$$\mathcal{K}^1(v^0(\tau)) = K_+ + K_- + 2C\cos(2\tau), \qquad \mathcal{T}^1(v^0(\tau)) = -4C\sin(2\tau). \tag{43}$$

The positive part has integral $4C$ on every period of length π. Artificial heat mixes the separated shear regions; the actual inviscid packets experience only constant translations.

Let v^δ be the actual unforced viscous evolution from $v_0 = v^0(0)$. It stays in the globally regular class

$$U^\delta_\tau = \delta U^\delta_{yy}, \qquad \Theta^\delta_\tau + U^\delta \Theta^\delta_x = \delta \Delta \Theta^\delta.$$

For a fixed sufficiently high Sobolev order, the heat estimate for $U^\delta - U$ and the differentiated passive-scalar energy estimate give

$$\left\| v^\delta(\tau) - v^0(\tau) \right\|_{H^s} \le C_s \delta (1+\tau)^2 e^{C_s \tau}, \qquad 0 < \delta \le 1. \tag{44}$$

Indeed $\left\| U^\delta - U \right\|_{H^{s+1}} \le \delta \tau \left\| U \right\|_{H^{s+3}}$. The scalar difference is transported by U^δ, whose spatial derivatives are bounded by those of U; its inhomogeneous terms are $-(U^\delta - U)\Theta^0_x + \delta \Delta \Theta^0$. The uniform Sobolev bounds of the periodic reference bound these terms by $C_s \delta (1+\tau)$, and Grönwall proves the displayed comparison. No unknown three-dimensional continuation estimate is used.

Set $\delta = \nu/\alpha$ and $u^\alpha(t) = \alpha v^{\nu/\alpha}(\alpha t)$. Choose a fixed $\gamma > 0$ with $C_s \gamma < 1/2$. On $0 \le \tau \le \gamma \log \alpha$, the comparison tends to zero uniformly as $\alpha \to \infty$. The ordered quintic kernel of $\mathcal{T}^1$ makes it locally Lipschitz on a sufficiently high Sobolev ball. Hence (43) still gives a positive-part integral bounded below by a positive constant times $\log \alpha$ on these actual viscous histories. Homogeneity gives

$$\mathcal{T}^\nu(\alpha v) = \frac{\alpha^5}{\nu^2} \mathcal{T}^1(v), \qquad \mathrm{d}t = \mathrm{d}\tau/\alpha,$$

which proves (40).

Finally, for any actual shear evolution $(F(y,t), 0, \Theta(x,y,t))$, put $M_0 = \left\| F(0) \right\|_\infty$ and $B_0 = M_0^2 \left\| \Theta(0) \right\|_2^2 / (2\nu^2)$. The maximum principle and scalar energy equality imply

$$\mathcal{K}^\nu(u(t)) + \int_0^t \frac{\left\| \mathcal{M}(u) \right\|_2^2}{\nu} \, \mathrm{d}s \le B_0.$$

To see this, write the horizontal stretching as $\mathrm{div}_{x,y}(Fe_x \otimes J\nabla\Theta)$, where $J\nabla\Theta = (\Theta_y, -\Theta_x)$, and use $\left\| \mathcal{M}(u) \right\|_2 \le M_0 \left\| \nabla \Theta \right\|_2$. More explicitly, the auxiliary heat energy identity and the actual scalar energy equality give the complementary bounds

$$\mathcal{K}^\nu(u(t)) \le \frac{M_0^2}{2\nu^2} \left\| \Theta(t) \right\|_2^2, \qquad \int_0^t \frac{\left\| \mathcal{M}(u) \right\|_2^2}{\nu} \, \mathrm{d}s \le \frac{M_0^2}{2\nu^2} \left(\left\| \Theta(0) \right\|_2^2 - \left\| \Theta(t) \right\|_2^2 \right).$$

The capacity identity and the complete positive balance therefore give, for every prefix,

$$-\mathcal{K}^\nu(u_0) \le \int_0^t \mathcal{T}^\nu \, \mathrm{d}s \le B_0 - \mathcal{K}^\nu(u_0), \qquad Y(u(t)) + \mathcal{K}^\nu(u(t)) + \int_0^t \mathcal{W}^\nu \, \mathrm{d}s \le Y(u_0) + B_0.$$

Here B_0 is quartic under amplitude scaling and $Y(u_0)$ is quadratic. This proves (41) and the ratio assertion. □

The lower bound uses a family of different data and a shrinking physical interval containing a growing number of turnover cycles. It is not infinite renewal on one fixed finite interval for one datum, and it is not a singularity. A general full-datum allowance can grow faster than quartically; the signed target is already paid in this example.

Corollary 14. *Use the periodic frequency separations above with $J = 0$, and fix $\eta, \zeta, \chi > 0$, with $\eta < 1/2$. For each datum αv_0 choose r, ρ, σ from its energy and viscosity so that the periodic versions of the three estimates give*

$$\|\mathcal{M}_{\mathrm{far}}\|_2 \le \eta\nu\sqrt{D}, \qquad |\mathcal{T}_{\mathrm{inner}}| \le \zeta\nu D, \qquad |\mathcal{T}_{\mathrm{pair}}| \le \chi\nu D,$$

and take $r \to \infty$ as $\alpha \to \infty$. Then the positive-part lower bound (40) also holds for $\mathcal{T}_{\mathrm{coupled}}$, after reducing its constant. Its signed primitive and $\int \mathcal{W}_{\mathrm{keep}}$ still have $O_\nu(\alpha^2 + \alpha^4)$ bounds on every prefix. The conservative lattice constants 200, 10^6 and 10^7, respectively, suffice for these choices of the three separations.

Proof. All nonzero velocity and nonlinear-direction modes have magnitude at least one, so $\mathcal{T}_{\mathrm{core}} = \mathcal{T}_{\mathrm{keep}}$. There is no low stretching output. The symmetric parent mask preserves the cancellation between two purely vertical inputs. Thus on v^0 its capacity has the form (43), with a masked cross pairing C_r. Absolute convergence of the smooth quartic Fourier formula gives $C_r \to C > 0$. The quintic kernel bound is uniform in the mask, so the viscous comparison remains uniform in r. Consequently $\int (\mathcal{T}_{\mathrm{keep}})_+$ has the same $\alpha^4 \log \alpha$ lower bound.

The two excluded returns have combined absolute integral at most $(\zeta+\chi)\int \nu D \le (\zeta+\chi)(Y(u_0)+B_0)$. Subtracting them cannot remove the logarithmic growth. Moreover,

$$\mathcal{W}_{\mathrm{keep}} \le 2\mathcal{W} + 2\eta^2\nu D, \qquad \mathcal{K}_{\mathrm{keep}} \le 2\mathcal{K} + \eta^2 Y.$$

The second inequality follows by integrating the far-source bound along the auxiliary heat flow. The modified positive identity, with $2|\langle Au, \mathcal{M}_{\mathrm{far}}\rangle| \le 2\eta\nu D$, then bounds the signed primitive of $\mathcal{T}_{\mathrm{keep}}$ by a constant times $Y(u_0) + B_0$. Subtract the two already paid returns. □

This corollary concerns the specified input/output floor and its data-selected separations. It neither rules out a separately chosen input cutoff nor refutes the signed remaining budget. It shows that cancellation over repeated history can remain necessary inside the coupled sector even after all three absolute sector payments.

9 Heat payment constrains changes of capacity

The recurrent example suggests measuring the actual instantaneous stretching instead of the stretching of heat-smoothed velocities. Such a change must also preserve, or explicitly replace, the viscous payment used in the positive balance. We continue on the normalized 2π torus with smooth mean-zero solenoidal fields.

Proposition 15. *Let $\mathcal{B}$ be a functional such that $s \mapsto \mathcal{B}(H_s u)$ is continuously differentiable and tends to zero as $s \to \infty$ for every such u. If*

$$-\frac{\mathrm{d}}{\mathrm{d}s}\mathcal{B}(H_s u) \ge \frac{1}{\nu}\,\|\mathcal{M}(H_s u)\|_2^2,$$

then $\mathcal{B}(u) \ge \mathcal{K}^\nu(u)$. If the heat derivative is exactly the displayed source cost, then $\mathcal{B} = \mathcal{K}^\nu$. In particular no functional depending only on $F(u) = (\operatorname{curl} u \cdot \nabla)u$, and vanishing when $F = 0$, can have this heat payment on the whole smooth class. The same conclusion holds with F replaced by $\mathcal{M}$.

Proof. Integrate the inequality from zero to S and let $S \to \infty$. The right side tends to (8). The equality case follows by the same argument. For the last assertions use the smooth datum v_0 of Proposition 13: its actual stretching and $\mathcal{M}$ vanish, whereas $\mathcal{K}^1(v_0) = K_+ + K_- + 2C > 0$. □

This statement concerns a specified heat payment and terminal limit. It does not exclude a larger parent reserve, a different positive cost, or an explicitly history-dependent functional.

For a concrete instantaneous choice, put

$$\mathcal{Q}^\nu(u) = \frac{1}{2\nu^2}\langle \mathcal{M}(u), A^{-1}\mathcal{M}(u)\rangle, \qquad \mathcal{G}_{\mathrm{out}} = (Y + \mathcal{Q}^\nu)/2.$$

The inverse is zero on the constant mode. This nonnegative capacity vanishes on the entire cyclic Euler reference above. Define

$$C_\nabla(u) = \sum_j (\partial_j \omega \cdot \nabla)\partial_j u, \qquad \mathcal{V}^\nu(u) = -\frac{2}{\nu}\langle A^{-1}\mathcal{M}(u), RC_\nabla(u)\rangle, \qquad \mathcal{T}^\nu_{\mathrm{out}} = D\mathcal{Q}^\nu(u)[N(u)].$$

The Laplacian product rule gives $D\mathcal{M}(u)[-\nu Au] = -\nu A\mathcal{M}(u) - 2\nu RC_\nabla(u)$. Consequently, along the actual unforced equation,

$$2\mathcal{G}'_{\mathrm{out}} + \mathcal{W}^\nu = \mathcal{T}^\nu_{\mathrm{out}} + \mathcal{V}^\nu. \tag{45}$$

The source and positive cost $\mathcal{W}^\nu$ are the original ones; $\mathcal{V}^\nu$ is an additional quartic term of undetermined sign.

Proposition 16. *Let*

$$u_* = (\sin y,\ 0,\ (-1 + \cos(2y))\sin x).$$

For every $\nu > 0$,

$$\mathcal{Q}^\nu(u_*) = \frac{67}{16000\nu^2}, \qquad \mathcal{K}^\nu(u_*) = \frac{23}{2400\nu^2}, \qquad D\mathcal{Q}^\nu(u_*)[-\nu Au_*] = \frac{49}{4000\nu} > 0.$$

Moreover $\Pi(u_*) = \mathcal{T}^\nu_{\mathrm{out}}(u_*) = 0$ and $\mathcal{V}^\nu(u_*) = 67/(2000\nu)$. For the globally regular unforced shear evolution from αu_*,

$$2\mathcal{G}'_{\mathrm{out}}(0) = -\frac{29}{2}\nu\alpha^2 + \frac{49}{4000\nu}\alpha^4.$$

This is positive when $\alpha^2 > 58000\nu^2/49$, although $Y'(0) = -29\nu\alpha^2/2 < 0$. In particular $\mathcal{V}^\nu \le \theta\mathcal{W}^\nu$ cannot hold at every smooth state for any $\theta \le 1$.

Proof. Writing s for auxiliary heat time, direct differentiation gives

$$\mathcal{M}(H_s u_*) = e_z \cos x \left[\left(\tfrac{1}{2}e^{-2\nu s} - \tfrac{1}{4}e^{-6\nu s}\right)\sin y - \tfrac{3}{20}e^{-6\nu s}\sin(3y)\right].$$

Orthogonality gives the two capacities, the displayed heat derivative, and $\|\mathcal{M}(u_*)\|_2^2 = 17/800$. The nonlinear velocity source is purely vertical and proportional to $\cos x$. Its contribution to $D\mathcal{M}$ is proportional to $\sin x$, so its pairing in $D\mathcal{Q}$ vanishes. The same x-orthogonality gives $\Pi = 0$. Finally $Y(u_*) = 9/4$ and $D(u_*) = 29/4$. The product-rule identity gives $\mathcal{V}^\nu = 49/(4000\nu) + 17/(800\nu)$. Homogeneity and (45) prove the remaining claims. For example, at $\nu = 1$, $\alpha = 40$ the exact values are

$$2\mathcal{G}'_{\mathrm{out}}(0) = 8160, \qquad \mathcal{W}^1(40u_*) = 77600, \qquad \mathcal{V}^1(40u_*) = 85760.$$

The triangular shear/passive-scalar equations provide global smoothness. Continuity turns the strict initial increase into an increase on a nonzero actual time interval; no uniform interval length is claimed. □

This rules out dropping, or automatically absorbing, the new viscous term. It does not rule out a signed physical-time allowance for it. A familiar larger parent reserve illustrates the remaining option:

$$\mathcal{B}_4(u) = \frac{\|u\|_4^4}{2\nu^2}, \qquad A_4 = \int |u|^2|\nabla u|^2, \qquad B_4 = \int \sum_j (u\cdot\partial_j u)^2.$$

Heat differentiation gives $D\mathcal{B}_4[-\nu Au] = -2(A_4+2B_4)/\nu$. Since $\|\mathcal{M}(u)\|_2^2 \le 2A_4$, this reserve has at least the required heat payment and hence $\mathcal{B}_4 \ge \mathcal{K}^\nu$. Its Euler derivative is instead

$$D\mathcal{B}_4(u)[N(u)] = \frac{2}{\nu^2}P_4(u), \qquad P_4(u) = -\int |u|^2 u\cdot\nabla p.$$

Pure transport cancels from this expression. The pressure-free shear cycles therefore disappear, but a signed pressure-work history bound is still required for arbitrary data. These comparisons identify the terms a changed capacity must control; they supply no such bound.

10 All-frequency renewal in a common polarized belt

The coherent part of the quintic return admits a direct estimate that retains the full generated velocity. Work either on $\mathbb{R}^3$ with the unitary Fourier convention used above or on the normalized 2π torus. Let v be a smooth solenoidal field with all Sobolev norms finite and Fourier support in $\{|k| \ge K\}$, where $K \ge 1$. Fix a real unit vector n and $\beta \ge 0$, and assume at every occupied frequency that

$$|k\cdot n| \le \beta|k|^{1/2}, \qquad |\widehat{v}(k)\cdot n| \le \beta|k|^{-1/2}|\widehat{v}(k)|. \tag{46}$$

The first restriction concerns frequency direction; the second concerns velocity polarization. Set $E_v = \|v\|_2^2$ and $D_v = \|\Delta v\|_2^2$. There is no restriction on the number of occupied shells.

Proposition 17. *With $C_* = 30\,000\,000\,000$, the full nonlinear renewal satisfies*

$$|\mathcal{T}^\nu(v)| \le \frac{C_*\beta^2(\beta+1)^{3/2}}{\nu^2}K^{-1/4}E_v^{3/2}D_v. \tag{47}$$

The same bound holds after multiplication by a scalar mask of modulus at most one in the paired five-input expansion used below. In particular, it permits the high stretching-output and nonlinear-output masks, and the symmetric separated-sector masks above. The generated velocity is $N(v)$, with its complete Leray projection; it is not projected into the input belt or its planar polarization.

Proof. For real nonzero frequencies p, q and transverse complex amplitudes a, b, write $\mathcal{S}(p,a;q,b)$ for the paired inverse-curl stretching symbol. Up to a common unit complex factor, it is

$$\frac{(p+q)\times((p\times q)\times(a\times b))}{|p+q|^2}.$$

At $p+q=0$ set this symbol to zero, as in $\mathcal{R}$. The identities behind the previously established polarized pair estimate give

$$|\mathcal{S}(p,a;q,b)| \le (8\delta+2\alpha+2\gamma)\min\{|p|,|q|\}|a||b|, \tag{48}$$

where

$$\delta = \max\{|p\cdot n|/|p|, |q\cdot n|/|q|\}, \quad \alpha = |a\cdot n|/|a|, \quad \gamma = |b\cdot n|/|b|.$$

Zero amplitudes are handled separately. To recall the algebra, when $\delta \le 1/2$, projection of a onto $n \times p$ has remainder at most $2|a \cdot n|$, and its scalar coefficient is at most $2|a|/|p|$. The corresponding planar pair is bounded by $8\delta|p \times q||a||b|$. Expanding the two remainders adds $(2\alpha + 2\gamma)|p \times q||a||b|$. Finally $|p \times q| \le |p + q| \min\{|p|, |q|\}$ cancels the inverse output. When $\delta > 1/2$, the elementary unpolarized bound $|\mathcal{S}| \le \min\{|p|, |q|\}|a||b|$ already implies (48). Thus no restriction on a generated mode's angle is hidden in this estimate.

For two original inputs satisfying (46), it follows that

$$|\mathcal{S}(p, a; q, b)| \le 12\beta\sqrt{\min\{|p|, |q|\}}|a||b|. \tag{49}$$

Next take two original inputs d, e with amplitudes w, z, let $r = d + e$, and let b_r be their ordered contribution to $N(v)$. Put $R = |r|$ and $L = \max\{|d|, |e|\}$. Transversality and the Leray projection give

$$\begin{aligned} |b_r| &\le R|w||z|, \\ |b_r \cdot n| &\le (\beta\sqrt{|e|} + |r \cdot n|)|w||z|, \qquad |r \cdot n| \le \beta(\sqrt{|d|} + \sqrt{|e|}). \end{aligned} \tag{50}$$

Indeed, before projection the axial component is the transport term $(w \cdot e)(z \cdot n)$, bounded by $\beta\sqrt{|e|}|w||z|$. The pressure correction is at most $|r \cdot n||w||z|$, using the output derivative in $w \cdot e = w \cdot r$. This keeps both transport of an existing axial component and axial velocity produced by pressure.

Let c be the remaining original input with amplitude a_c, and put $S = \max\{|p|, |q|, |c|\}$. Frequency conservation in the quintic pairing is $p + q = c + d + e$, hence $R \le 3S$. Applying (48) to (c, a_c) and (r, b_r) gives

$$\begin{aligned} |\mathcal{S}(c, a_c; r, b_r)| &\le \beta \min\{|c|, R\} \left(\frac{10R}{\sqrt{|c|}} + 22\sqrt{L} \right) |a_c||w||z| \\ &\le \beta(30S^{3/2} + 22S\sqrt{L})|a_c||w||z|. \end{aligned}$$

The case $r = 0$ contributes zero. The heat integration uses precisely

$$\nu^2(|p|^2 + |q|^2 + |c|^2 + |r|^2)$$

in the denominator. The two original parents d, e inside N are not independently heat-evolved. Pairing the first two inputs replaces twice their ordered source by $\mathcal{S}(p, a; q, b)$, while the two derivative slots give $\mathcal{S}(c, a_c; r, b_r)$. Consequently, the absolute quintic kernel, after removing the five amplitude norms, is bounded by

$$\frac{\beta^2}{\nu^2} \left(360 + 264\sqrt{L/S} \right). \tag{51}$$

This pairing must precede any mask that could distinguish the two orders. Every mask asserted in the proposition respects that pairing.

For $R_0 \ge 1$, the original support inside $|k| < R_0$ has at most $32(\beta + 1)R_0^{5/2}$ lattice points. To see this uniformly in n, place disjoint unit cubes at those points. They lie inside a cylinder of radius $2R_0$ and half-height $(\beta + 1)\sqrt{R_0}$, whose volume is less than the stated bound. The same bound dominates the normalized Fourier volume in $\mathbb{R}^3$. Thus the Fourier L^1 mass, including one convolution normalization on $\mathbb{R}^3$, satisfies

$$L_1(R_0) \le [32(\beta + 1)]^{1/2} R_0^{5/4} \sqrt{E_v}. \tag{52}$$

First consider $L \le 8S$, where (51) is at most $1200\beta^2/\nu^2$. Let j be the largest dyadic index of the five original frequencies. The second largest magnitude is at least a quarter of the largest, by their signed sum being zero. Both therefore belong to

$$B_j = \{2^{j-2} \le |k| < 2^{j+1}\}, \qquad E_j = \int_{B_j} |\widehat{v}(k)|^2 \, \mathrm{d}k,$$

with the integral replaced by a sum on the torus. There are at most ten choices of these two positions. With the other three frequencies fixed, Cauchy–Schwarz for the two large inputs costs E_j. The three remaining integrals cost $L_1(2^{j+1})^3$. Since $32^{3/2} < 192$ and $2^{15/4} < 16$, this region is bounded by

$$\frac{10 \cdot 1200 \cdot 192 \cdot 16}{\nu^2} \beta^2 (\beta+1)^{3/2} E_v^{3/2} \sum_j 2^{15j/4} E_j.$$

Each frequency occurs in at most three bands, and $2^{15j/4} \le 256|k|^{15/4}$ on B_j. Hence the final sum is at most $768K^{-1/4}D_v$.

In the remaining region $L > 8S$, let $j = \lfloor \log_2 L \rfloor$ and $m = \lfloor \log_2 S \rfloor$. Then $m \le j-3$. Both inner parents belong to $B'_j = \{2^{j-1} \le |k| < 2^{j+1}\}$, because $|r| \le 3S < 3L/8$. Write E'_j for the energy in this band. Now (51) is at most $900\beta^2\nu^{-2}2^{(j-m)/2}$. Cauchy–Schwarz costs E'_j, and the three other factors cost $L_1(2^{m+1})^3$. Their sum is bounded by

$$\frac{900 \cdot 192 \cdot 16}{\nu^2} \beta^2 (\beta+1)^{3/2} E_v^{3/2} \sum_j E'_j 2^{j/2} \sum_{m \le j-3} 2^{13m/4}.$$

The inner geometric sum is less than $2\,2^{13j/4}$. Also $\sum_j 2^{15j/4} E'_j \le 32K^{-1/4}D_v$ by two-band overlap. The two resulting numerical constants sum to

$$10 \cdot 1200 \cdot 192 \cdot 16 \cdot 768 + 900 \cdot 192 \cdot 16 \cdot 2 \cdot 32 = 28\,488\,499\,200 < C_*.$$

This proves (47). Every estimate uses a nonnegative majorant after pairing, so admissible masks do not enlarge it. Finite frequency restrictions and these summable majorants justify the full expansion by convergence; no bound depends on the number of shells. □

For $E_v \le M$, choosing the fixed cutoff so that

$$C_* \beta^2 (\beta+1)^{3/2} M^{3/2} K^{-1/4} \le \theta \nu^3 \tag{53}$$

pays this entire return by $\theta\nu D_v$. This is compatible with arbitrary rotation of the common axis between time slices: the estimate does not differentiate the axis. It requires one axis for all of the coherent inputs at a given time, and does not cover a different axis at each scale or a spatially varying family of axes.

The estimate can be used without assuming preservation of Fourier support. At a fixed time, split the actual high field as $v = w + d$, where w is an orthogonal Fourier selection satisfying (46). One may instead select belt frequencies and project their velocities orthogonally onto $n \times k$, which gives exact planar polarization. Expand the actual quintic return by multilinearity. The term with all five original inputs from w is bounded by (47), with $E_w \le E_v$ and $D_w \le D_v$. This term still includes $N(w)$ in full, including modes outside the selected belt. Every remaining term has at least one original input from d. A measurable time-dependent selection creates no derivative term in this algebraic partition of the instantaneous return; no evolution equation or separate capacity is asserted for w.

There is no estimate here for those remaining terms, for multiple belts, or for maintaining a small defect under unforced evolution. Thus the result pays a coherent interaction class in the signed-history target; it does not supply the target for arbitrary data. In particular, replacing $N(w)$ by its planar part would be invalid: the finite example in the companion checker changes the renewal from positive to negative under that replacement.

11 Different axes in adjacent shells

The preceding estimate cannot be extended by assigning an independent axis to each shell. The obstruction is an interaction between two families whose individual stretching is identically zero.

Proposition 18. *Fix $\nu > 0$. For arbitrarily large dyadic λ, there are real solenoidal Schwartz data $u_{0,\lambda} = a_\lambda + b_\lambda$ on $\mathbb{R}^3$, with $\|u_{0,\lambda}\|_2^2 = 1$, such that*

$$\begin{aligned} &\operatorname{supp} \widehat{a}_\lambda \subset \{2\lambda < |k| < 4\lambda,\ |k_3| \le \sqrt{\lambda}\}, && (a_\lambda)_3 = 0, \\ &\operatorname{supp} \widehat{b}_\lambda \subset \{4\lambda < |k| < 8\lambda,\ |k_2| \le \sqrt{\lambda}\}, && (b_\lambda)_2 = 0. \end{aligned} \tag{54}$$

Thus each occupied dyadic shell has one belt satisfying (46) with $\beta = 1$ and exact planar polarization, but the two axes differ. Let $Q_\lambda = 2\lambda$. There are actual smooth unforced solutions from these data on $[0, t_\lambda]$, where $t_\lambda \asymp \lambda^{-15/4}$, and constants $c, C > 0$ independent of λ, such that

$$\int_0^{t_\lambda} \mathcal{T}^\nu_{Q_\lambda, Q_\lambda}(u_\lambda)\, \mathrm{d}t \ge c\, \nu^{-2} \lambda^{1/2}, \tag{55}$$

$$\int_0^{t_\lambda} \mathcal{W}^\nu(u_\lambda)\, \mathrm{d}t \le C(\nu + \nu^{-1}) \lambda^{1/4}. \tag{56}$$

In particular their excess over any fixed multiple of the full cost and any allowance fixed by unit energy, viscosity and horizon is unbounded. The cutoff varies with the datum.

Proof. A mixed coefficient. On the normalized 2π torus, let A have coefficients $i(-k_2, k_1, 0)$ at

$$(3,1,0), \quad (2,1,0), \quad (1,2,0),$$

and let B have coefficients $i(k_3, 0, -k_1)$ at

$$(5,0,4), \quad (-1,0,-4),$$

in each case including the reality partners. These are two-dimensional flows in different planes, so $F(A) = F(B) = 0$. The complete convolution and parent heat integration nevertheless give

$$\mathcal{T}^1_{2,2}(sA + tB) = s^3 t^2 c_0, \qquad c_0 = \frac{5469265097437563}{230433883895600} > 0. \tag{57}$$

Here the input cutoff removes no parent. Both internal high-output masks also retain the coefficient; removing them leaves the same value. For clarity about the degrees, each B frequency has third coordinate ± 4, so a resonant quintic tuple has an even number of B inputs. Four B inputs would leave one A input with a nonzero second coordinate. Pure A terms have zero stretching. Thus only three A and two B inputs can contribute. The finite rational calculation in (57) includes their actual generated velocity and both differentiated parent slots.

Two localized plane profiles. Choose nonzero nonnegative Schwartz functions χ, g on $\mathbb{R}$ with compact Fourier support in $[-1,1]$, and normalize $g(0)=1$. Let $\psi_A(x,y)$ and $\psi_B(x,z)$ be twice the sums of the cosines at the respective frequencies above. Set

$$U_L(x,y) = (-\partial_y, \partial_x, 0)[\chi(x/L)\chi(y/L)\psi_A(x,y)],$$
$$V_L(x,z) = (\partial_z, 0, -\partial_x)[\chi(x/L)\chi(z/L)\psi_B(x,z)].$$

For large L, their plane Fourier supports lie strictly inside the annuli $(2,4)$ and $(4,8)$. For small $\delta > 0$ define the Schwartz field

$$W_\delta(x,y,z) = U_L(x,y)g(\delta z) + V_L(x,z)g(\delta y). \tag{58}$$

Both summands are exactly solenoidal and retain zero velocity component in their own normal direction. Their disjoint radial spectra give the exact identity

$$\|W_\delta\|_2^2 = q/\delta, \qquad q = \|g\|_{L^2(\mathbb{R})}^2 \left(\|U_L\|_{L^2(\mathbb{R}^2)}^2 + \|V_L\|_{L^2(\mathbb{R}^2)}^2\right) > 0. \tag{59}$$

For each fixed integer m, $\|W_\delta\|_{H^m} \le C_m \delta^{-1/2}$. No whole-space L^2 assertion is made about the limiting unmodulated plane fields.

We claim that, for a suitable fixed L,

$$\mathcal{T}^1_{2,2}(W_\delta) \longrightarrow c_L > 0 \quad \text{as } \delta \downarrow 0. \tag{60}$$

Here is the packet justification, retaining the different dimensions of the two families. Expand the paired quintic expression by the original input family. A pure-family term has a kernel of order δ^2, by the two polarized pair bounds and the generated axial estimate used in Proposition 17. Its three Fourier L^1 factors are bounded, and the two L^2 factors cost $O(\delta^{-1})$. Thus its contribution is $O(\delta)$.

For a mixed term, write the original frequencies as $(x_i, y_i, \delta z_i)$ for the first family and $(x_i, \delta y_i, z_i)$ for the second. Each normal Fourier density has a factor δ^{-1} canceled by its integration measure. Eliminate the three frequency constraints using an x coordinate, a native y coordinate from the first family, and a native z coordinate from the second. The corresponding determinant is one in absolute value. All remaining profile variables have compact support. The retained internal frequencies are bounded away from zero by the cutoff 2, and the heat denominator contains three original frequencies bounded away from zero. The quintic kernel is uniformly bounded. Its sharp masks converge off sets of measure zero. Dominated convergence therefore gives a finite plane-frequency integral, while the five normal profiles contribute $g(0)^5 = 1$. This defines c_L and proves existence of the limit.

Finally let $L \to \infty$ in this finite plane-frequency integral. There are finitely many carrier tuples. Nonresonant tuples vanish for large L. In a resonant tuple, rescale the ten plane offset coordinates by L; the three frequency constraints contribute L^3. Every carrier output is separated from the cutoff 2. The retained symbols converge to the finite symbols in (57). The differentiated envelopes contribute errors of order L^{-1} in this normalization. Only the degree-three/degree-two mixed terms survive, so

$$L^{-3} c_L \longrightarrow c_0 \left(\int_{\mathbb{R}} \chi^5\right)\left(\int_{\mathbb{R}} \chi^3\right)\left(\int_{\mathbb{R}} \chi^2\right) > 0. \tag{61}$$

The common x envelope occurs five times, the first family's y envelope three times, and the second family's z envelope twice. Fix L large enough to obtain (60).

We also have, for this fixed L,

$$\|\mathcal{M}(W_\delta)\|_2 \le C. \tag{62}$$

For either pure family the paired inverse-curl symbol is $O(\delta)$, its Fourier L^1 mass is bounded, and its L^2 norm is $O(\delta^{-1/2})$, giving an $O(\delta^{1/2})$ bound. For the mixed term write $F_i = \sum_j \partial_j(\omega_j u_i)$ and apply the multipliers $\mathcal{R}\partial_j$, which are bounded on L^2. Products of a Schwartz (x, y) profile and a Schwartz (x, z) profile, including their derivatives, have uniformly bounded whole-space L^2 norms. The normal envelopes and their derivatives are uniformly bounded for $0 < \delta \le 1$. This proves (62) without spending the divergent L^2 norm of an entire plane profile.

An actual short interval. Let $w^{\delta,\epsilon}$ solve the unforced equation with initial datum W_δ and viscosity $0 \le \epsilon \le 1$. The usual high-Sobolev local estimate, discarding the nonpositive viscous term, gives a time $c\delta^{1/2}$ on which

$$\left\|w^{\delta,\epsilon}\right\|_{H^4} \le C\delta^{-1/2}, \qquad \left\|\partial_\tau w^{\delta,\epsilon}\right\|_{H^2} \le C\delta^{-1}.$$

These bounds are uniform in ϵ. The bounded quartic heat kernel and the continuous quadratic map $N : H^2 \to L^2$ give

$$|D\mathcal{T}^1_{2,2}(v)[h]| \le C\,\|v\|^4_{H^2}\,\|h\|_{H^2}\,.$$

All three fixed cutoff operators have norm at most one on the Sobolev spaces used here. Also $\|\mathcal{M}(v) - \mathcal{M}(z)\|_2 \le C(\|v\|_{H^2} + \|z\|_{H^2})\,\|v - z\|_{H^2}$. It follows that

$$\left|\frac{\mathrm{d}}{\mathrm{d}\tau}\mathcal{T}^1_{2,2}(w^{\delta,\epsilon})\right| \le C\delta^{-3}, \qquad \left\|\mathcal{M}(w^{\delta,\epsilon}(\tau)) - \mathcal{M}(W_\delta)\right\|_2 \le C\tau\delta^{-3/2}.$$

By (60), choose $c_1 > 0$ small, independently of δ, ϵ. On $0 \le \tau \le c_1\delta^3$ these estimates give

$$\mathcal{T}^1_{2,2}(w^{\delta,\epsilon}) \ge c > 0, \qquad \left\|\mathcal{M}(w^{\delta,\epsilon})\right\|_2 \le C, \qquad \left\|\Delta w^{\delta,\epsilon}\right\|_2^2 \le C\delta^{-1}. \tag{63}$$

This is a deliberately short uniform interval; no turnover-time persistence or invariant spectral family is assumed.

Concentration at fixed physical viscosity. For large dyadic λ put

$$\delta = \lambda^{-1/2}, \qquad a = \lambda^{5/4}/\sqrt{q}, \qquad \epsilon = \nu\lambda/a = \nu\sqrt{q}\,\delta^{1/2}, \qquad u_\lambda(t,x) = a\,w^{\delta,\epsilon}(a\lambda t, \lambda x).$$

For large λ, $\epsilon \le 1$. This solves the actual unforced equation with viscosity ν. Equation (59) gives unit initial energy, and the exact support and polarization give (54). Set $t_\lambda = c_1\delta^3/(a\lambda) = c_1\sqrt{q}\,\lambda^{-15/4}$. The full scaling formulas are

$$\int_0^{t_\lambda} \mathcal{T}^\nu_{2\lambda,2\lambda}(u_\lambda)\,\mathrm{d}t = \frac{a^4\lambda^{-3}}{\nu^2}\int_0^{c_1\delta^3} \mathcal{T}^1_{2,2}(w^{\delta,\epsilon})\,\mathrm{d}\tau,$$
$$\int_0^{t_\lambda} \mathcal{W}^\nu(u_\lambda)\,\mathrm{d}t = \frac{a^3\lambda^{-2}}{\nu}\int_0^{c_1\delta^3} \left[\epsilon^2\left\|\Delta w^{\delta,\epsilon}\right\|_2^2 + \left\|\mathcal{M}(w^{\delta,\epsilon}) - \epsilon(-\Delta)w^{\delta,\epsilon}\right\|_2^2\right]\mathrm{d}\tau.$$

Since $\epsilon^2\delta^{-1} = \nu^2 q$, the bracket in the second integral is at most $C(1+\nu^2)$ by (63). The two prefactors times δ^3 are respectively $q^{-2}\nu^{-2}\lambda^{1/2}$ and $q^{-3/2}\nu^{-1}\lambda^{1/4}$. This proves (55)–(56). □

At the initial time the same construction has $\mathcal{T}^\nu_{Q_\lambda,Q_\lambda}(u_{0,\lambda}) \ge c\nu^{-2}\lambda^{17/4}$ and $\|\Delta u_{0,\lambda}\|_2^2 \le C\lambda^4$. Thus even an instantaneous bound $|\mathcal{T}^\nu_{Q,Q}| \le C\nu^{-2}E^{3/2}D$ with no decaying cutoff factor fails for the independently oriented shells. Replacing δ by a fixed positive multiple of $\lambda^{-1/2}$ makes both belt constants arbitrarily small and preserves the same powers of λ.

The initial geometry in (54) is not claimed to persist throughout the interval. New modes are generated by the complete equation. The construction does not refute a history bound that

separately assumes and uses maintained geometry, a full-datum allowance, or a different cutoff chosen for that datum. Indeed, the initial gradient energy lies between $4\lambda^2$ and $64\lambda^2$. The finite burst above is compatible with payment from that larger initial resource. No infinite recurrence, global smoothness of this family, or finite-time singularity is proved. The result locates a mixed interaction that the common-axis payment does not cover.

12 Verification boundary

The Lean theorem `openAI_solution_no_uniform_signed_production_budget` applies to the actual construction at every positive viscosity and concerns the full-field spatial cubic integral. The sharp Fourier-tail extension and all caloric results here are analytic proofs, not Lean theorems or axioms. The companion `labs/caloric_force.py` checks the complete caloric balance by exact rational Fourier arithmetic on the periodic cube, including forces with a gradient part. It also checks the three-part frequency split, including a nonzero low-output contribution, and the ordered mode bound on nearly cancelling parents. Separate source-output checks retain the original positive cost and include parent frequency parameters up to one million that feed low stretching outputs. These checks do not certify the whole-space estimates or a physical-time bound. The checks `labs/generated_caloric_renewal.py` and `labs/thin_caloric_renewal.py` reproduce the finite coefficients used in Proposition 9. `labs/caloric_localization.py` checks the strict cutoff gap, the complete scaling factors and a finite periodic streamfunction envelope. Its rational calculations retain the envelope derivative and all generated modes. They do not certify the Schwartz localization limit or the common local PDE time interval, which are analytic arguments above. The rational checker `labs/separated_caloric.py` tests the Schur matrix bounds, enlarged-band overlap, exact source partition, heat translation, and (36) with the full actual source. An independent quartic polarization checks the masked derivative. It also verifies the unchanged positive generated-child coefficient. Its finite periodic source tests use the conservative lattice constant 200, not the whole-space constant 10: counting at most $125\,2^{3j}$ lattice points in Ω_j replaces $2\sqrt{17}$ by $20\sqrt{85} < 200$ in the proof. The infinite-shell estimate and its whole-space use remain analytic. The checker `labs/inner_caloric.py` expands all five original inputs and verifies that their separated and remaining parts add to the independently evaluated derivative. It keeps the correct heat rate of the nonlinear output. A family with cancelling parent frequencies 128, 8192 and 2097152 gives the same positive separated return $166548224/(167049575\,\nu^2)$; reversing the velocity reverses its sign. The test therefore checks a nonzero payment, not just a vanishing sector. Its norm inequality uses a conservative periodic constant 10^6 in place of the whole-space 20. Counting a ball by $(3R_m)^3$ gives constant $(272/15)\,6\cdot 81\sqrt{3}\,2^{11/2} < 10^6$ in the same argument. The general estimate remains the analytic proof above. The checker `labs/paired_caloric.py` retains both original parent-shell labels inside N and tests the disjoint three-part partition against the complete derivative. Its inner part also agrees with the independently grouped calculation. Sign reversal and rescaling the two high amplitudes check the signs and degree of the selected terms. Its finite torus norm checks use constant 10^7 instead of the whole-space 100; the same lattice counting gives $80\cdot 81\sqrt{3}\,2^{9/2}(136/7) < 10^7$. The checker verifies that the earlier positive generated-child coefficient remains after all these separations. The checker `labs/shear_caloric_cycles.py` verifies the rational constants in the recurrent shear construction and finite heat pairings, including their phase dependence, viscosity scaling and behavior under the symmetric parent mask. It also checks the cancellation between two purely vertical inputs before applying that mask. Independent physical products, inverse curl and heat quadrature agree with those pairings. The finite Fejér profiles are explicitly checked to have a nonzero plateau-equation residual: they test the heat formula, not the

smooth Euler reference. The flat-plateau construction, logarithmic-time viscous comparison and all-prefix bounds are analytic proofs, not Lean certificates or numerical PDE time enclosures. The checker `labs/instantaneous_stretching.py` verifies the alternative capacity's complete Laplacian product rule and unforced positive balance, the exact shear constants, amplitude and viscosity scaling, and cases with a nonzero nonlinear return. It compares derivatives from both parent slots with an independent quartic polarization. Physical products and Fourier differentiation on two grids independently recover the shear values. These finite checks do not certify the heat-minimality argument or a general signed history estimate. The checker `labs/belt_caloric.py` expands the paired five-input return with its actual generated modes and heat rate. Exact rational arithmetic compares it with the existing parent derivative and an independent quartic polarization, including both internal output masks. The examples include nearly cancelling roots and widely separated inner parents, nonzero initial axial velocity, generated axial pressure, and both signs of renewal. One example has positive full renewal but negative renewal after either planarizing the generated mode or using its parents' heat rates. Exact checks also cover the norm inequality on these finite fixtures and the constants in the dyadic sum; a separate floating-point ratio samples the intermediate kernel bound. The all-frequency estimate, uniform lattice count and geometric decomposition remain analytic proofs. The checker `labs/crossed_belt_caloric.py` verifies the positive two-plane coefficient with exact rational arithmetic, classifies every original input by its plane, and checks the resulting cubic/quadratic amplitude degrees. Both internal high-output masks, the full quadratic source, an independent quartic polarization, and viscosity factors are retained. Finite normal and common-coordinate envelopes check the predicted resonant moments, including derivatives of the latter envelope. They are periodic Fourier calculations, not approximations certified in whole-space norms. Separate rational identities check the concentration, time and complete-cost scaling. The transverse-plane localization, uniform local interval and resulting whole-space obstruction are analytic, not Lean-certified or numerical PDE enclosures. Arbitrary-data unforced regularity remains open.

References

[1] OpenAI, *Finite time blowup for Navier–Stokes*, manuscript, with formalization in NavierStokesAndEuler, 2026.